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A unified rodent atlas reveals the cellular complexity and evolutionary divergence of the dorsal vagal complex

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eLife Assessment

This manuscript applies state-of-the-art techniques to define the cellular composition of the dorsal vagal complex in two rodent species (mice and rats). The result is a **fundamental** resource that substantially advances our understanding of the dorsal vagal complex's role in the regulation of feeding and metabolism while also highlighting key differences between species. The analyses of single-cell profiling experiments in the manuscript provide **compelling** insight into the cellular architecture of the dorsal vagal complex, with potential implications for obesity therapeutics.

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Abstract

The dorsal vagal complex (DVC) is a region in the brainstem comprised of an intricate network of specialized cells responsible for sensing and propagating many appetite-related cues. Understanding the dynamics controlling appetite requires deeply exploring the cell types and transitory states harbored in this brain site. We generated a multi-species DVC cell atlas using single nuclei RNAseq (sn-RNAseq), by curating and harmonizing mouse and rat data, which includes >180,000 cells and 123 cell identities at 5 granularities of cellular resolution. We report unique DVC features such as *Kcnj3* expression in Ca^{+}-permeable astrocytes as well as new cell populations like neurons co-expressing *Th* and *Cck*, and a leptin receptor-expressing neuron population in the rat area postrema which is marked by expression of the progenitor marker, *Pdgfra*. In summary, our findings demonstrate a high degree of complexity within the DVC and provide a valuable tool for the study of this metabolic center.

1. Introduction

Regulating appetite in response to changing environmental conditions to maintain energy balance requires the intricate interplay of multiple organs. Across the central nervous system, multiple discrete brain regions and cell types control appetite[1]. Within the brainstem, the dorsal vagal complex (DVC), comprising the area postrema (AP), nucleus of the solitary tract (NTS) and dorsal motor nucleus of the vagus (DMV), acts as a key nexus point for metabolic signals, evidenced by the dense expression of many relevant hormone receptors[2]. Separately, the DVC also serves as the primary site of gut-based signals via vagal afferents[3–5]. Functionally, the DVC was long considered a site promoting meal termination[6] but growing evidence suggests it also controls long-term energy balance[7].

Much of the interest in DVC neurons' role in appetite and energy balance stems from their role as therapeutic targets for obesity and anorexia in cancer cachexia[8, 9]. For many years, most pharmacotherapies failed to produce efficacious and long-lasting weight loss in obesity and increase appetite in cancer cachexia[10–12]. More recently, however, the glucagon-like peptide 1 receptor (GLP1R) agonists are eclipsing the weight loss achieved by previous generations of therapeutics[13]. While expressed across the central nervous system, GLP1R expression is particularly enriched in DVC neurons[14] and it is the DVC that mediates many of the effects of GLP1R agonists[11]. Separately, an emerging strategy for treating cancer cachexia is inhibiting growth differentiation factor 15 (GDF15) signaling in the DVC by blocking the action of GDNF Family Receptor Alpha Like (GFRAL)[15], the receptor for GDF15[16]. Given the clinical significance of the DVC for this spectrum of conditions, understanding the molecular heterogeneity and function of this region is essential.

Traditionally, identification and ensuing study of DVC cell types was limited to known neurotransmitter and receptor-expressing cell types. For instance, the cholecystokinin and monoamine-expressing cells, demarcated by tyrosine hydroxylase (*Th*) expression, were previously shown to be non-overlapping cell populations with unique roles in appetite suppression[17]. More recently, the advent of single-cell mRNA sequencing approaches made unbiased molecular censuses of heterogenous populations possible. Such studies focusing on the mouse have revealed the complexity of the mouse DVC, comprised of dozens of transcriptionally distinct cell types[2, 18, 19]. Moving forward, unifying information from multiple single-nuclei RNA sequencing (snRNA-seq) datasets of the DVC into a reference atlas is needed to more fully understand the cells within the DVC and generate a common and scalable classification for cell identities. Such a reference atlas is also needed to contrast the cell types found within the mouse to other research models, including the rat, which are prominent models for DVC biology[2, 20].

To address this, we generated a comprehensive snRNA-seq based atlas of the mouse and rat DVC. This atlas features mouse and rat labelled cells with increasing granularity and an accompanying computational toolbox including the mouse and rodent atlas label transfer environment using treeArches. The mouse DVC atlas reveals a higher degree of cellular complexity than previously appreciated. The cellular census of the rat DVC exhibits many similarities to the mouse but also rat-specific cells including a novel population of leptin receptor and *Pdgfra*-expressing neurons localized to the AP. Using this unified atlas, we also uncover the cell types that transcriptionally respond to meal consumption which may hold the potential of suppressing appetite without gastrointestinal side effects.

2. Materials and methods

2.1. Experimental design

To generate a snRNA-Seq based atlas of the mouse dorsal vagal complex (DVC), we isolated the DVC from 30 adult C57BL/6J mice and subjected them to 10x Genomics-based single-nuclei RNA barcoding and sequencing. Mice were either fasted overnight (n=5), fasted overnight and then refed 2 hours prior to euthanasia (n=10), fed ad libitum (n=5), injected with LiCl (n=5) or injected with vehicle (n=5). Conserved cell clusters across these groups were then identified.

2.2. Animals

All animals were bred in the Unit for Laboratory Animal Medicine at the University of Michigan and the Research Institute of the McGill University Health Centre. Procedures performed were approved by the University of Michigan Committee on the Use and Care of Animals, in accordance with Association for the Assessment and Approval of Laboratory Animal Care and National Institutes of Health guidelines. Alternatively, procedures were approved by the animal care committee of McGill University.

2.2.1. Mouse tissue for sn-RNA sequencing assay

Wild-type C57BL/6J mice (Jackson laboratories) were given ad libitum access to food (Purina Lab Diet 5001) and water in temperature-controlled (22°C) rooms on a 12-hour light-dark cycle with daily health checks. Mice were euthanized by isoflurane anesthesia and decapitated, then the brain was removed from the skull and aligned in a brain matrix. The cerebellar cortex was removed and the DVC was micro-dissected and flash frozen in liquid nitrogen (Fig.S1). To limit any circadian-driven gene expression changes, all animals were euthanized 4 hours following the onset of the light phase. Frozen tissue was stored at -80°C until use.

For assessment of feeding-related gene expression, mice were either ad libitum fed or food was removed at the onset of the dark cycle and the animals were fasted overnight. Animals were then euthanized the next morning under ad libitum fed or fasting conditions. A subset of fasted animals (n=5) were provided food and allowed to refeed for 2 hours prior to euthanasia.

2.2.2. Mouse and rat tissue collection for *in-situ* hybridization and immunofluorescent assays

Animals were provided with ad libitum access to food (Purina Lab Diet 5001) and water in temperature-controlled (22°C) rooms on a 12 h light-dark cycle with daily health status checks. C57BL/6J mice (Jackson laboratories) or Sprague Dawley rats (Jackson laboratoties) were euthanized by isoflurane anesthesia followed by CO_2 asphyxiation. Animals were then trans-cardially perfused with phosphate buffered saline for 3 minutes followed by 5 minutes perfusion with 10% formalin. Brain tissue was then collected and postfixed for 24 hours in 10% formalin before transfer into 30% sucrose for a minimum of 24 hours. Brains were then sectioned as 30 µm thick, free-floating sections.

2.2.3. Rat tissue collection for snRNA sequencing assays

Wild-type Sprague Dawley rats (Charles River) were provided with ad libitum access to food (Purina Lab Diet 5001) and water in temperature-controlled (22°C) rooms on a 12-hour light-dark cycle with daily health checks. Rats were euthanized by intraperitoneal injections of pentobarbital and decapitated, then the brain was removed from the skull and aligned in a brain matrix. The cerebellar cortex was removed and the DVC was dissected and flash frozen in liquid nitrogen. Frozen tissue was stored at -80°C until use.

2.3 Isolation of nuclei from DVC tissue

Frozen tissue was pooled (5 pooled DVC from mice and 2 DVCs from rats) and homogenized in Lysis Buffer (EZ Prep Nuclei Kit, Sigma-Aldrich) with Protector RNAase Inhibitor (Sigma-Aldrich) and filtered through a 30mm MACS strainer (Miltenyi). Filtered samples were centrifuged at 500 rcf for 5 minutes at 4°C, and pelleted nuclei were resuspended in fresh-made wash buffer (10 mM Tris Buffer, pH 8.0, 5 mM KCl, 12.5 mM MgCl2, 1% BSA with RNAse inhibitor) before undergoing a second filtration and centrifugation. The pelleted nuclei were resuspended in wash buffer with propidium iodide (Sigma-Aldrich) and underwent fluorescence-activated cell sorting (FACS) on a MoFlo Astrios Cell Sorter to remove debris. PI+ nuclei were collected and centrifuged at 100 rcf for 5 minutes at 4°C and resuspended in wash buffer to obtain a concentration of 750-1,200 nuclei/µL in preparation for sequencing.

2.4. Single-nuclei RNA-sequencing

Library preparation was performed by the Advanced Genomics Core at the University of Michigan. RT mix was added to target approximately 10,000 nuclei recovered per sample and loaded onto the 10× Chromium Controller chip. The Chromium Single Cell 3' Library and Gel Bead Kit v3, Chromium Chip B Single Cell kit, and Chromium i7 Multiplex Kit were used for subsequent RT, cDNA amplification, and library preparation, as instructed by the manufacturer. Libraries were sequenced on an Illumina NovaSeq 6000 (pair ended with read lengths of 150 nt).

2.5. Processing of the snRNA-sequencing files

From the snRNA-seq, we obtained four FASTQ files (two per sequencing lane, containing forward and reverse sequences respectively) per mouse treatment and two files with the forward and reverse sequences for the rat samples. We pre-processed the murine FASTQ files using either CellRanger 5.0.1 with inclusion of introns or CellRanger 7.0.1 with default parameters (which include intronic sequences) to align sequencing reads to the murine genome[21]. We aligned to the *Mus musculus* refdata-gex-mm10-2020-A reference genome. For the rat files, we used CellRanger 3.0.1 with no introns inclusion and aligned using the *Rattus norvegicus* genome assembly Rnor_6.0[21]. We obtained 110,167 and 13,360 cells from the murine and rat experiments, respectively. In addition to these two datasets, we also used the following steps to process the murine Dowsett dataset[18] publicly available as a raw gene expression matrix set of files. We processed the gene expression matrices in R[22] with Seurat v5 and SeuratObject v5 to remove low quality cells[23]. For processing and analysis, R v4.3.1[22] was used. We filtered cells with ≥500 number of genes mapped and >1.22 RNA unique molecular identifiers (UMIs) counts/genes mapped ratio (based on the first 2 percentiles per sample) (Table S1). Subsequently, we merged the data per treatment, calculated the median RNA content of clusters at resolution 1.0 and those below the first quartile of the median distribution were removed (Table S2). Additionally, we used DoubletFinder v2.0.3 based on the expected doublets rates by 10X Genomics after adjusting for homotypic doublets modeled as the sum of squared annotation frequencies[24] (Table S3). On each iteration we processed the data with libraries scaled to 10,000 UMIs per cell and log-normalized. We identified the most variable genes computing a bin Z-score for dispersion based on 20 bins average expression. We regressed UMI counts and used principal component (PC) analysis for dimensionality reduction on to the top 2,000 most variable genes. We used the first 30 PCs for *k*-nearest neighbors clustering and for uniform manifold approximation and projection (UMAP) projections using Seurat v5[23] default parameters. Murine samples integration in a single matrix was done with Harmony[25], after which clustering and UMAP projections were performed using the harmony embeddings instead of PCs.

2.6. Mapping of our murine data to existing brain databases

We mapped the resulting murine integrated high quality singlets to the celldex v1.12.0 built-in mouse RNA sequencing reference and other four mouse databases[26–29] (Table S4) using SingleR v2.4.1[30] and scRNA-seq v2.16.0[31] packages in R[22]. Since each database has its own cell type nomenclature, in our murine dataset we established a 'likely-cell type' to be unanimous among all databases, for example if all databases labeled a cell as 'neuron'. We initially labeled the cells with conflicting or unassigned label by one or more databases as 'unknown'. To confirm the cell type of all the cells labeled and to identify the unknown cell identities, we mapped 473 markers from the literature. For this, the average expression per marker gene was obtained on scaled data for each cluster at resolution 1.0 (i.e. 48 clusters) (Table S5). We also manually visualized UMAP projections from all these markers in our dataset. Some clusters (e.g. cluster 27) showed in the previous step to contain cells belonging to different cell types, so we subset and reprocessed these clusters using the Harmony embeddings to assign their cell identity based on marker expression at the lowest resolution level that allowed to separate them.

2.7. Assigning high resolution clustering-based cell identity in the murine dataset

In the murine dataset, for all non-neuronal clusters at resolution 1.0 gene expression markers were obtained using the FindConservedMarkers function from Seurat v5[23] package in R[22]. The function was run using the MAST algorithm[32] on scaled data with a log fold-change (FC) threshold of 0.25. Only upregulated marker genes detected on 40% of all cells in that cluster, were retained. Since the FindConservedMarkers gives a log2FC output per sample, we calculated the average log2FC across samples and the proportion of samples for which a gene had an adjusted p-value ≤0.05. A marker gene was considered as such if it had an average log2FC>1 across samples and was significant (i.e. had an adjusted p-value ≤0.05) in more than 80% of the samples. Furthermore, neurons were subset, reprocessed and subjected to new clustering through Seurat v5[23]. We obtained the gene markers per cluster at resolution 1.0 as described. We named the clusters based on the known functions of the upregulated/downregulated genes (e.g. myelinating oligodendrocytes), peculiarities of the cell groups (e.g. Ca^{+}-permeable astrocytes) or their upregulated gene expression markers if the previous methods were unsuitable (e.g. Ano2 neurons) (Table S6). All visualizations, unless otherwise stated, were done using ggplot2 v3.5.0 and ComplexHeatmap v2.18.0 packages in R[22].

2.8. Oligodendrocytes trajectory inference

We subset the oligodendrocytes (i.e. pre-myelinating, myelinating intermediate and myelinating) in our murine dataset and placed each cell on an inferred cellular trajectory using their transcriptomic data in R[22]. We used the SCORPIUS v1.0.9 package[33] with default parameters to perform dimensionality reduction by PCA on log-normalized counts and to obtain the trajectory image, with a random seed of 1,000. We additionally performed the analysis on the Harmony embeddings from the original data integration.

2.9. Gene expression files format interconversions

To convert Seurat ('RDS') DVC datasets to 'h5ad' format and vice versa, we used reticulate v1.35.0[34] and anndata 0.7.5.6[35] packages in R[22] with import of scipy, scanpy, numpy and anndata modules from Python v3.9[36]. To convert .txt, .csv and .tsv datasets (i.e. the Ludwig dataset[2], the spinal cord dataset[37] and the cortex/hippocampus database[38]), we converted those to 'h5ad' format in Python v3.9[36].

2.10. Murine DVC hierarchy construction

We converted our murine labeled count matrix to h5ad. We used treeArches[39] in Python v3.9[36] to create a manual tree using the three layers of cellular granularity in our database. We reprocessed the samples using the treeArches pipeline[39] to normalize the count data (counts normalized per cell and then log-normalized). After multiple tests, we determined that the 2,500 most variable genes was the optimum parameter for the integration of other datasets based on our murine data, therefore we identified the top 2,500 most variable genes and integrated the samples using scVI and treeArches[39, 40]. This reference latent space obtained after integration was used to generate the UMAP embeddings. This sample integration was done to ensure that inter-sample variations were removed for the cell identity steps. We trained our manual tree based on the cell identity layer-3 labels using default parameters.

We subset the Ludwig dataset[2] to match the initial 2,500 most variable genes from our datasets and performed surgery to incorporate labels from the Ludwig dataset[2]. The two layers of granularity established by Ludwig[2] were combined in the variable “identity_layer3” to contain the highest granularity of cell identity for neurons and non-neuronal cell types. Next, we normalized the count data as specified for our murine dataset, and mapped the Ludwig dataset on the reference latent space using scArches. We then used the learn_tree function with default parameters for hierarchical progressive learning from scHPL v1.0.5[41] which yielded a hierarchy of harmonized cell labels from both datasets. We printed all trees using matplotlib v3.8.2[42] and scHPL v1.0.5[41]. All steps performed and scripts used in this phase are available in the GitHub repository: *https://github.com/LabSabatini/DVC_cell_atlas*.

2.10.1. Corroboration of the validity of our murine cell hierarchy using treeArches prediction modality

Using treeArches[39] and its dependencies in Python v3.9[36] we compared the original cell labels in our and Ludwig datasets with the resulting prediction of the label for each cell using our learned hierarchy. We used the predict_labels function from scHPL v1.0.5[41] with default parameters, and the latent representation obtained when we constructed our murine hierarchy. We compared the original and the predicted labels through a heatmap visualization. Next, we predicted the labels of new data, a murine dataset by Dowsett which comprises the NTS, a part of the DVC[18]. We processed the Dowsett data similarly to the Ludwig dataset and obtained the query latent representation. We then used the predict_labels function from scHPL v1.0.5[41] with default parameters to transfer the labels in our hierarchy to the Dowsett dataset based on gene expression similarity among the 2,500 initial most variable genes. We then compared the resulting labels from this prediction to the manual mapping of our cell identities based on maker expression (Table S7) using UMAP embeddings in R[22].

2.11. Labeling of our rat DVC dataset

We processed our rat dataset as previously described to obtain high-quality singlets. Based on the three layers of cell identity that we created to label the murine data from our experiments, we calculated the average expression of the top 5 marker genes for each of the layer-3 identities (except duplicated genes and mouse-specific genes) on every rat cluster to identify those that were equivalent and could be labeled as the murine data (**Tables S6** and **S8**). Since this dataset is smaller than the mouse data, we decided to use clusters at resolution of 2.0 to capture more heterogeneity and delineate better the non-neuronal cell populations. Clusters 27 and 29 were further subset and reclustered since we found them to contain a mix of endothelial cells, mural cells, tanycytes and fibroblasts based on marker expression. Microglia and mixed neurons were sub-clustered as well. The lowest resolution level that allowed to separate populations was used to label the cells. Neurons were also subset and we mapped the murine neuronal classes markers to resolution 2.0 markers which were manually curated to delineate better the different populations within them. For all cell identities, and because we only applied one treatment to the rats (i.e. fed ad libitum), we used the FindMarkers function from Seurat v5[23] package in R[22] to obtain gene expression markers of each rat cell population. The function was run using the MAST algorithm[32] with same parameters used for the murine data (Table S9).

2.12. Construction of the rodent DVC hierarchy

We then integrated our rat labeled dataset to the murine samples in Python v3.9[36] using the initial top 2,500 most variable genes initially defined in our mouse data. Incorporation of rat labels was done as previously described for the Ludwig dataset using the learn_tree function for hierarchical progressive learning from scHPL v1.0.5[41]. The cell populations that were not rejected but placed at the bottom of the tree were moved manually to their belonging parent branches. For example, the 'smooth_muscle_rat' label was moved to be a children node of the mural cells node. If the parent branch did not exist, we created it. For example, the 'immunity_related_rat' label belonged to a new rat neuronal class, therefore, we incorporated this parent node within the neurons node and then we moved the 'immunity_related_rat' label to be a children node. After manual curation, we re-trained the tree. We printed all trees using matplotlib v3.8.2[42] and scHPL v1.0.5[41]. All steps performed and scripts used in this phase are available at the GitHub repository *https://github.com/LabSabatini/DVC_cell_atlas* for reproducibility.

2.13. Mapping gene expression in other brain-site snRNA-seq datasets

We downloaded labeled datasets from six brain regions different from the DVC: the HypoMap, spinal cord, cortex, hippocampus, pons and forebrain databases[37, 38, 43, 44]. For the HypoMap data, we subset the database by randomly selecting cells belonging to 32 included hypothalamic samples. We processed the datasets in Seurat v5[23] as previously described and obtained Harmony[25] embeddings to obtain UMAP projections. The labels used in the visualization are the original curated labels provided on each dataset. Neurons were further subset from the processed datasets and reprocessed as previously described. The other datasets were processed in the same manner after subsetting astrocytes. The pons and forebrain data was downloaded using scRNAseq v2.16.0 packages in R through SingleR v2.4.1[30].

2.14. Differential gene expression analysis in murine treatments

We subset the murine dataset neurons and glial cells at layer 2 granularity of identity and obtained differential gene expression for pair-wise comparison of treatments (i.e. refed versus ad libitum fed, refed versus fasted and LiCl versus vehicle injected mice). We used the FindMarkers function from Seurat v5[23] package in R[22] using the MAST algorithm[32] on log-normalized counts with a minimal log fold-change threshold of 0.1 and default parameters. The number of cells per treatment can be found in Table S10.

2.15. Immunofluorescent staining

Free-floating brain sections from mouse and rat were washed with PBS, three times for five minutes. The sections were then blocked for one hour in PBS containing 0.1% Triton X-100 and 3% normal donkey serum (Fisher Scientific). The sections were incubated overnight at room temperature with Goat anti-PDGFRA (Invitrogen, CAT#, 1:200), HuC/HuD (Invitrogen, A-21271, 1:30). The following day, sections were washed and incubated for two hours with Alexflour488 and Cy3 conjugated secondary antibody (Jackson immunoresearch, 1:500). Tissues were then washed three times in PBS before being mounted onto glass slides, covered in mounting media (Fluoromount-G, Southern Biotechnology) and coverslipped. Imaging was performed using an

Olympus BX61 microscope and a Zeiss LSM780-NLO laser scanning confocal microscope equipped with IR-OPO lasers at the Molecular Imaging Platform, Research Institute of the McGill University Health Centre (RI-MUHC), Montreal, CA.

2.16. *In-situ* hybridization and imaging

Brain sections containing the DVC from four wild-type C57BL/6J mice obtained as described above, were fixed on glass slides and stored at -20°C for no more than 36 hours. We followed the manufacturer's ACD 323100 user manual for the RNAscope® Multiplex Fluorescent Reagent Kit v2 Assay for fixed frozen tissue samples. Briefly, we rinsed the slides with 1X PBS and incubated them at room temperature for 10 minutes after adding ~1-2 drops of H_2O_2 per section. We removed the H_2O_2 from slides and rinsed them twice with distilled water (diH_2O). Next, we submerged the slides in 200 mL of hot diH_2O for 10 seconds followed by 200mL of RNAscope® 1X Target Retrieval Reagent for 5 minutes, using a steamer with lid. After briefly transferring the slides to room temperature diH_2O, we submerged them in 100% ethanol for 3 minutes and allowed them to air dry. We applied approximately 1-2 drops of RNAscope® Protease III to each section and incubated them at 40°C for 30 minutes in a HybEZ^{TM} oven with diH_2O wet paper in the tray. After rinsing the slides with diH_2O, we hybridized the probes at 40°C for 2 hours.

From this point on, we performed all steps by incubating the slides at 40°C in the HybEZ^{TM} oven with humid tray followed by rinse using RNAscope® 1X Wash Buffer. We applied approximately 1-2 drops of RNAscope® reagents AMP1, AMP2, AMP3 and then intercalated HRP channel (15 minutes), fluorophore (30 minutes) and HRP blocker (15 minutes) for each channel present in the probe mix. The mix of probes used for the *Th*/*Cck* co-expression assay was RNAscope® Mm-Th-C2 (317621-C2) and Mm-Cck-C3 (402271-C3) diluted in probe diluent as specified in the user manual. For *Hcrt* we used RNAscope® Mm-Hcrt-C2 (490461-C2) and for *Agrp*, Mm-Agrp (400711), in a dilution as specified in the user manual. The used fluorophores were Cy3 for *Cck* and *Hcrt* (1:1,500), and Cy5 for *Th* and *Agrp* (1:1,000) diluted in RNAscope® TSA buffer. DAPI dye (1:1,000) was applied at the end of the assay, and the samples were stored for ~48 hours at -4°C, protected from light, before imaging. We imaged the sections using a Zeiss LSM780-NLO laser scanning confocal microscope with IR-OPO lasers at the Molecular Imaging Platform at the RI-MUHC, Montreal, CA.

3. Results

3.1. The murine dorsal vagal complex cell classes

To generate a *de novo* snRNA-seq atlas of the mouse DVC, we isolated the DVC from 30 adult C57BL/6J mice and subjected them to 10X Genomics-based snRNA barcoding and sequencing (**Fig.1A**). From the CellRanger pre-processing pipeline[21] we obtained 110,167 cells of which 99,740 were retained after processing. Cell identities were initially mapped to 5 databases from different brain regions and then curated manually using 473 cell identity markers (**Tables S4-S5**). We distinguished clusters of neurons and glial, vascular and connective tissue cells, a first layer of cell identity (**Fig.1B**; Table S9). We further describe a second layer of cellular resolution, which are subgroups of the first layer that contain the cell classes (**Fig.1C**; Table S7).

Among the ten neuronal cell classes identified at layer 2 resolution, we found two major groups which we called 'magnaclass1' and 'magnaclass2' (Fig.S2). Magnaclass1 neurons predominately express genes required for inhibitory neurotransmission (e.g. *Gad1*, *Gad2*) and share transcriptomic markers with the Sall3-class of neurons, whereas magnaclass2 neurons express mostly excitatory transcripts (e.g. the glutamate transporter VGLUT2) also found in the Lhx2-class and monoamine-class (**Fig.1D-E**; Fig.S2). However, as described in other brain areas[45, 46], evidence of co-transmission/co-release of multiple primary neurotransmitters was found in at least 55% of neurons regardless of their cell class (**Fig.1D-E**; Table S11). We also identified 8

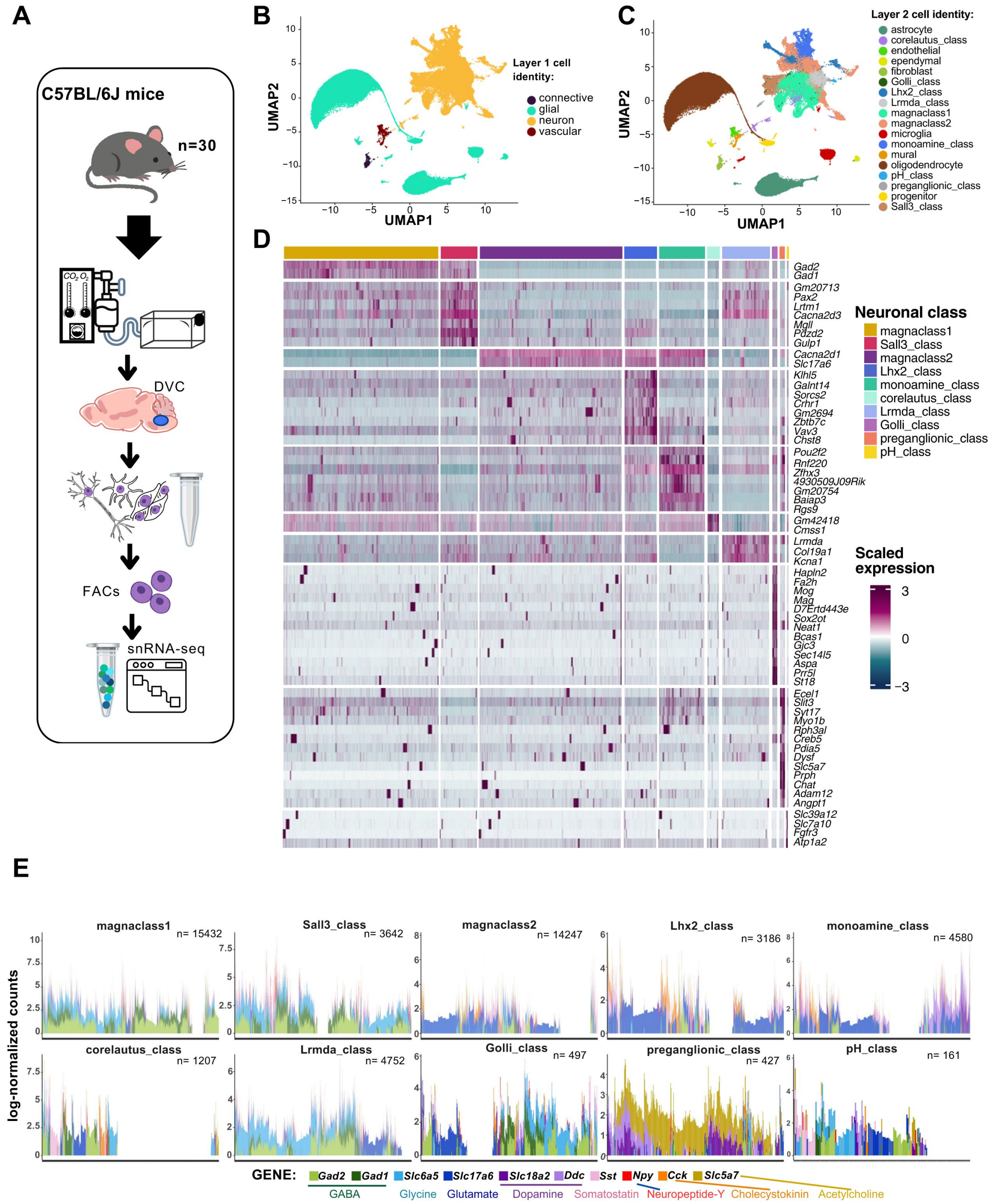

Figure 1

Two layers of cell identity for the murine DVC cells from snRNA-seq data

A. Adult C57BL/6J mice were subjected to 10X Genomics-based snRNA barcoding and sequencing after dissection of their DVC (n=30). **B.** Labeled UMAP plot of the mouse DVC (cells n=99,740) at the lowest resolution (layer 1) including 4 cell identities, and **C.** the cell class (layer 2) including 18. **D.** Heatmap of the main marker genes (average log2FC expression >80 percentiles of upregulated genes) for each neuronal class of the mouse DVC (MAST algorithm; neurons n= 48,131; adj. p<0.05). Each column is one cell and each row is one marker gene. **E.** Stacked barplot of log-normalized counts of genes for 10 transcripts (i.e. *Gad2*, *Gad1*, *Slc6a5*, *Slc17a6*, *Slc18a2*, *Addc*, *Sst*, *Npy*, *Cck* and *Slc5a7*) related to transmission/release of 8 neurotransmitters and neuropeptides, by neuronal cell class. Each bar represents one neuron (n=48,131).
snRNA-seq=single-nuclei RNA-sequencing; DVC=dorsal vagal complex; FACs=fluorescence-activated cell sorting; UMAP=uniform manifold approximation and projection; FC=fold-change; adj.=adjusted; GABA=gamma-aminobutyric acid

transcriptionally distinct neural classes in addition to the magnaclasses. Among these 8 classes we found two neural classes expressing traditionally non-neuronal genes; the Golli and pH-related classes. The Golli-class neurons are a relatively small cluster, comprising ~1% of all neurons. Interestingly, Golli-class neurons express multiple myelin-associated genes such as *Mag* and *Mog*[47–49] normally expressed by oligodendrocytes (**Fig.1D**; Table S6). Additionally, we identified a class of neurons with genes normally upregulated in astrocytes (e.g. *Slc6a11*) and related to clearance and uptake of glutamate, gamma-aminobutyric acid (GABA), Ca^+ and bicarbonate [50, 51], which we named 'pH-related class' (Table S6). Similar to Golli-class neurons, pH-related neurons were a relatively rare cluster (<1% of all neurons). While the corelautus-class neurons express very few genes at higher levels, among them are *Cdk8*, *Lars2*, *Gabra6* and *Fat2*. Expression of *Cdk8* is largely absent from the DVC but is enriched within the DVC-adjacent hypoglossal nucleus (Fig.S3) suggesting these cells are hypoglossal neurons. We also identified the Lhx2-Class which was enriched for the Parvalbumin (*Pvalb*) gene whose expression is largely excluded from the DVC. As *Pvalb* is highly expressed in the neighbouring cuneate nucleus (Fig.S3), we posit this class may belong to this anatomically adjacent area which was captured in our dissection. When interrogating snRNA-seq data from the hypothalamus[43] (Fig.S4), another brain site important for energy balance, we could not find clusters sharing all markers with our neuronal classes, indicating DVC-specific neuronal programs (Fig.S4C).

3.2. The murine DVC cells at their highest resolution

To better define the heterogeneity of the mouse DVC, we further assigned a clustering-based third layer of cellular resolution resulting in fifty cell identities, of which 15 are non-neuronal (**Fig.2A**). At this resolution we differentiated between two major groups of astrocytes. Differential expression in synapse-related factors revealed a group of astrocytes with high expression of the inwardly rectifying GIRK1 (*Kcnj3*) channel[52] and lack of expression of the AMPAr subunit GluA2 (*Gria2*), additionally rendering them K^+/Na^+/Ca^+-permeable[53, 54] (**Fig.2B**). We speculate these astrocytes are specific to the DVC as we could not find *Kcnj3+*; *Gria2*-astrocytes in the hypothalamus, forebrain, cortex or hippocampus[38, 43, 44] (Fig.S4D; Fig.S5). Furthermore, although some regions close to the DVC such as pons and the spinal cord show some co-expression of *Kcnj3* and *Gfap*, the patterns are not to the same extent as observed in the DVC [44] (Fig.S5F-G). In addition, we distinguished a spectrum of DVC mature oligodendrocytes, including a group resembling the intermediate oligodendrocytes described in the other brain sites that decrease in number with age[37, 55] (**Fig.2C**). This gradient of expression of markers including *Fyn*, *Opalin* and *Anln* in oligodendrocytes (**Fig.2C**), do not respond to a developmental trajectory (Fig.S6). Although 'myelinating-intermediate' and 'myelinating' oligodendrocytes include cells with overlapping gene expression programs, there is no evidence that each oligodendrocyte will follow a single trajectory (Fig.S6). Furthermore, to properly separate some cell identities, we performed sub-clustering in three DVC clusters (i.e. clusters 23, 26 and 27) (Fig.S7A). This permitted us to label pre-myelinating oligodendrocytes, endothelial and mural cells, and distinguish two fibroblast sub-types (Fig.S7). One of these subtypes expresses high *Stk39* previously described in fibroblast stress-response[56] and so was termed 's-fibroblasts' for stress-associated fibroblasts (Fig.S7B), which we failed to detect in the hypothalamus (Fig.S4E). Additionally, we excluded the possibility of monocytes in our microglial data (Fig.S8A), which had no evidence of microglial activation as we mapped minimal levels of the activation genes *Cxcl10*, *Cd5*, *Cxcl9* and *Zbp1*[57], therefore we posit that our DVC microglial cells are in a basal state (Fig.S8B). To further evaluate possible activation states in microglia, we performed sub-clustering and evaluated their marker genes (Fig.S8C). We have identified a group of microglia expressing oligodendrocyte markers and resembling a subpopulation of hippocampal microglia in an Alzheimer's disease model which is disease-protective and synaptic-function-supporting[58], and a cortical/subcortical phagocytic subpopulation in a multiple sclerosis model[59]. A second group expresses *Ndrg2* and receptors for GABA and glutamate which have been related to microglia that may induce neuronal damage[60, 61] (Fig.S8D).

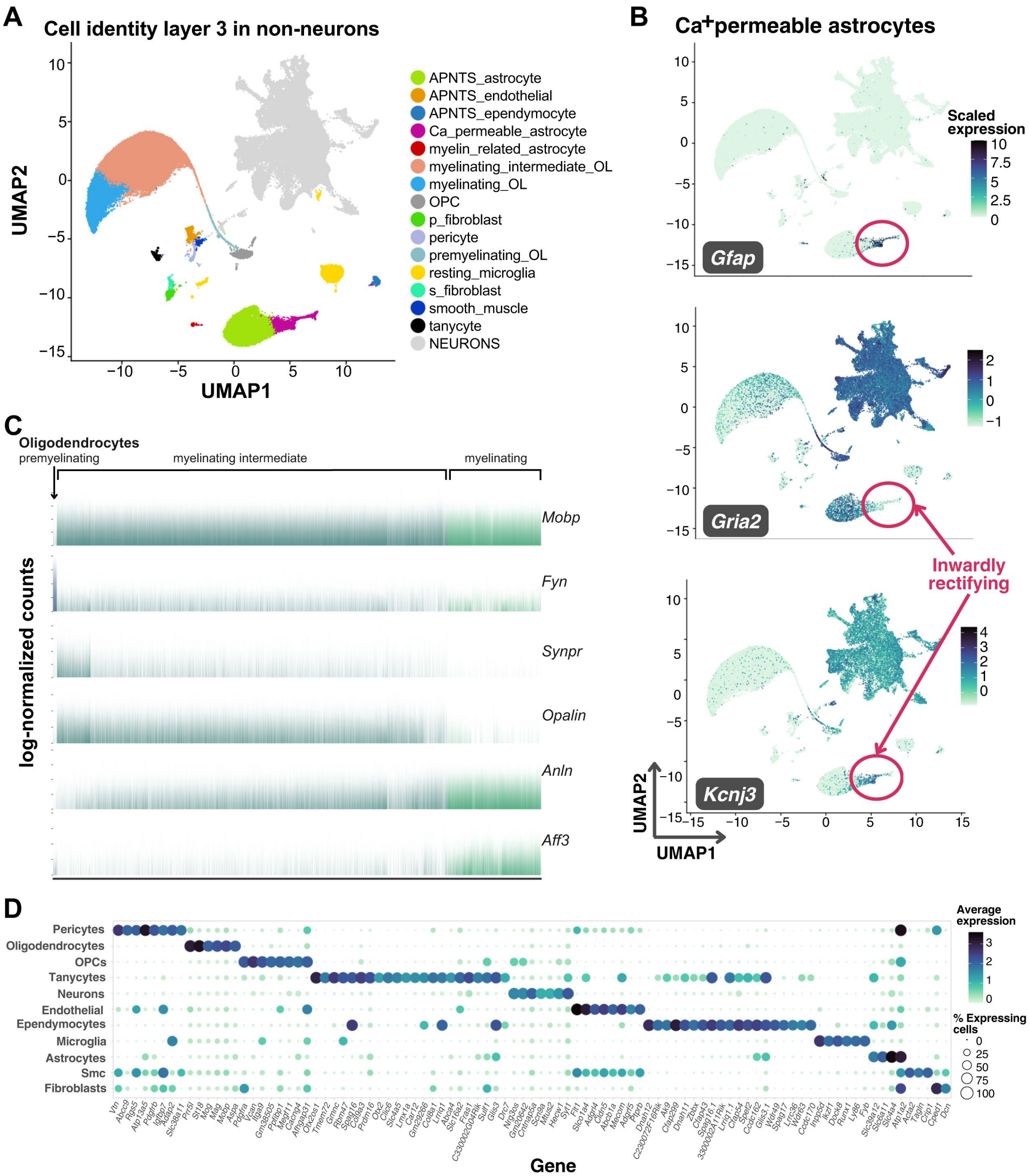

Figure 2

Third layer of non-neuronal murine DVC cell identities

A. Labeled UMAP plot of the mouse DVC with the 15 non-neuronal cell identities shown (total cells n=99,740; non-neuronal cells n=51,609). **B**. UMAP plot showing scaled expression of *Gfap*, *Gria2* and *Kcnj3* genes in Ca^{+}-permeable astrocytes (circled; total cells n=99,740; Ca^{+}-permeable astrocytes n=1,719). **C.** Barplot of log-normalized expression of six genes in oligodendrocytes (premyelinating n=267; myelinating intermediate n=25,692; myelinating n= 6,170). Only premyelinating oligodendrocytes lack expression of *Mobp*, involved in myelin formation. Reduction of *Synpr* and *Opalin* expression is observed as *Anln* and *Aff3* increase in myelinating oligodendrocytes. **D.** Balloon plot of the main cell types of non-neuronal cells and neurons showing average log-normalized counts of their marker genes (MAST algorithm; adj. p<0.05). Markers shown are upregulated genes in >80% of cells per group with average log2FC>4, or upregulated in >70% of cells with average log2FC>8.

DVC=dorsal vagal complex; UMAP=uniform manifold approximation and projection; APNTS=Area postrema and nucleus of the solitary tract; OL=oligodendrocyte; OPC=oligodendrocyte precursor cell; Smc=smooth muscle cells

We also obtained the DVC gene expression markers of the main cell types in the central nervous system and neurons (**Fig.2D**). To more thoroughly classify DVC neural types, we subset and re-clustered the 10 neural classes from layer 2 which resulted in 35 neuronal identities as a third layer of granularity (**Fig.3A**; Table S6). These identities include 'preganglionic' or *Chat*-expressing neurons and a large number of neurons with unspecific markers which we called 'mixed neurons' (Table S6). Of the mixed neuron classes, two are largely excitatory (Mixed neurons 3 and 4) and two largely inhibitory, however, cells in the four mixed neuron classes share similar transcriptional profiles with many other neural classes. When we performed sub-clustering on mixed neurons, we could distinguish a total of 10 subtypes from the original 4 mixed neuronal identities (Fig.S9); however, transcriptional differences between these were subtle. Interestingly, a mixed neurons-like group of cells was previously described in the murine hypothalamus[26], suggesting multiple brain sites may contain neurons which lack highly differentiable transcriptional programs from snRNA-sequencing data.

Additionally, we split the monoamine-class into four layer-3 cell identities (i.e. M0, M1, M2 and M3) with specific cell markers (**Fig.3B**; Table S6). M0 and M1 neurons express genes for enzymes synthesizing norepinephrine and serotonin, but M0 expresses minimal levels of the transporter VMAT2 (*Slc18a2*) (**Fig.3C**; Table S6). Due to the low proportion of cells expressing *Th* and *Tph2* in M2 and M3 neurons, we consider them monoamine-modulators as they seem to synthesize trace amines (**Fig.3C**; Table S6). Of note, the receptor for GDF15, GFRAL, is expressed heavily by a subset of M0 neurons, some of which co-express the calcium sensing receptor (*Casr*) gene (**Fig.3C**).

We named 'GLP1' a neuronal cluster with high expression of the pre-proglucagon gene (*Gcg*) (**Fig.3D**; Table S6), which also expresses high levels of both the prolactin and leptin receptors[62] (**Fig.3E**). Additionally, these cells have the highest proportion of DVC neuronal co-expression of glutamate- and GABA-related genes (**Fig.3F**); mainly *Gad2* with considerably lower expression of *Slc32a1*.

In our analysis we found a subset of monoaminergic neurons that express the cholecystokinin (*Cck*) gene (**Fig.4A**). While the *Cck* and *Th*-expressing DVC neurons are considered to be non-overlapping cell types with distinct physiologies[17, 63], but we quantified >10% of the *Th*-expressing neurons also expressing *Cck* in our snRNA-seq data. We further confirmed co-expression of *Th* and *Cck* results by *in-situ* hybridization (**Fig.4B**). The majority of this overlap occurs in cells belonging to the COE-Atp10a and monoamine class (i.e. M0, M3) cell identities (**Fig.4A**).

3.3. The murine DVC cell atlas

To generate a comprehensive murine DVC database from multiple datasets, we utilized treeArches[39] to harmonize our snRNA-seq DVC labels with that from a publication by Ludwig and collaborators (i.e. the 'Ludwig dataset')[2] (**Fig.5A-B**). After building our initial cell hierarchy (Fig.S10), we integrated our dataset with the Ludwig dataset, giving a total of 171,868 cells (**Fig.5B**). Next, the labeled cell identities from the Ludwig dataset[2] were incorporated into our tree based on the similarity between the transcriptomic programs of our labeled identities and the Ludwig-labeled identities using progressive learning through scHPL[41] (**Fig.5A**). Some of the Ludwig dataset identities failed to be incorporated into our tree, like the oligodendrocyte precursor cells (OPCs), which contain a mix of OPCs and pre-myelinating oligodendrocytes (Fig.S11A). The mix of cell programs makes it impossible for treeArches to correctly add one branch of the Ludwig OPCs to the tree as ideally it would be added as a subpopulation of both the OPCs and pre-myelinating oligodendrocytes, which is not biologically possible. The resultant murine hierarchy has a fourth layer of cellular resolution as some of Ludwig identities, mainly neurons, are subgroups of our cell identities (Fig.S11B).

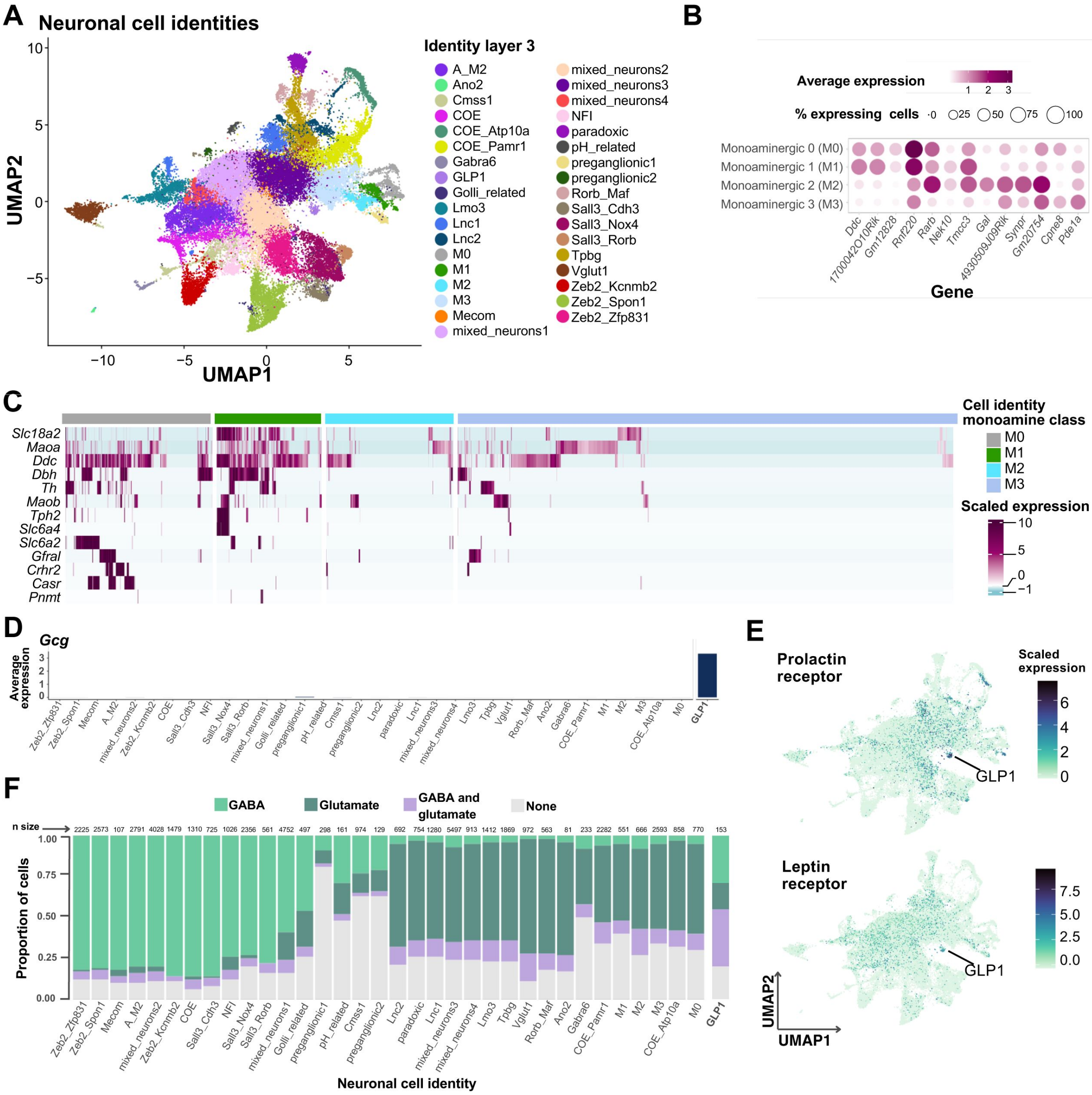

Figure 3

Neuronal populations of the murine DVC

A. Labeled UMAP plot with the neuronal layer 3 cell identities (neurons n=48,131). **B**. Monoamine class balloon plot showing average log-normalized expression of marker genes for each cell identity (MAST algorithm; M0 cells n=770, M1 cells n=551, M2 cells n=666 and M3 cells n=2,593; adj. p<0.05), and **C.** heatmap of expression of monoamine related genes. In the heatmap, each column represents one cell and each row, one gene. **D**. Barplot of *Gcg* average log-normalized expression by neuronal cell identity. E. UMAP plot showing the expression of leptin receptor and prolactin receptor genes in neurons. The GLP1 cluster is highlighted (GLP1 cells n=153). **F**. Stacked barplot of the proportion of cells expressing one or more GABA-related genes (i.e. *Slc32a1*, *Gad1*, *Gad2*) and glutamate-related genes (i.e. *Slc17a6*, *Slc17a7*). The proportion of cells per cell group co-expressing GABA and glutamate associated genes are shown in purple. Only cells with log-normalized counts >0 were considered to express the genes.
DVC=dorsal vagal complex; UMAP=uniform manifold approximation and projection; Res.=resolution; GABA=gamma-aminobutyric acid

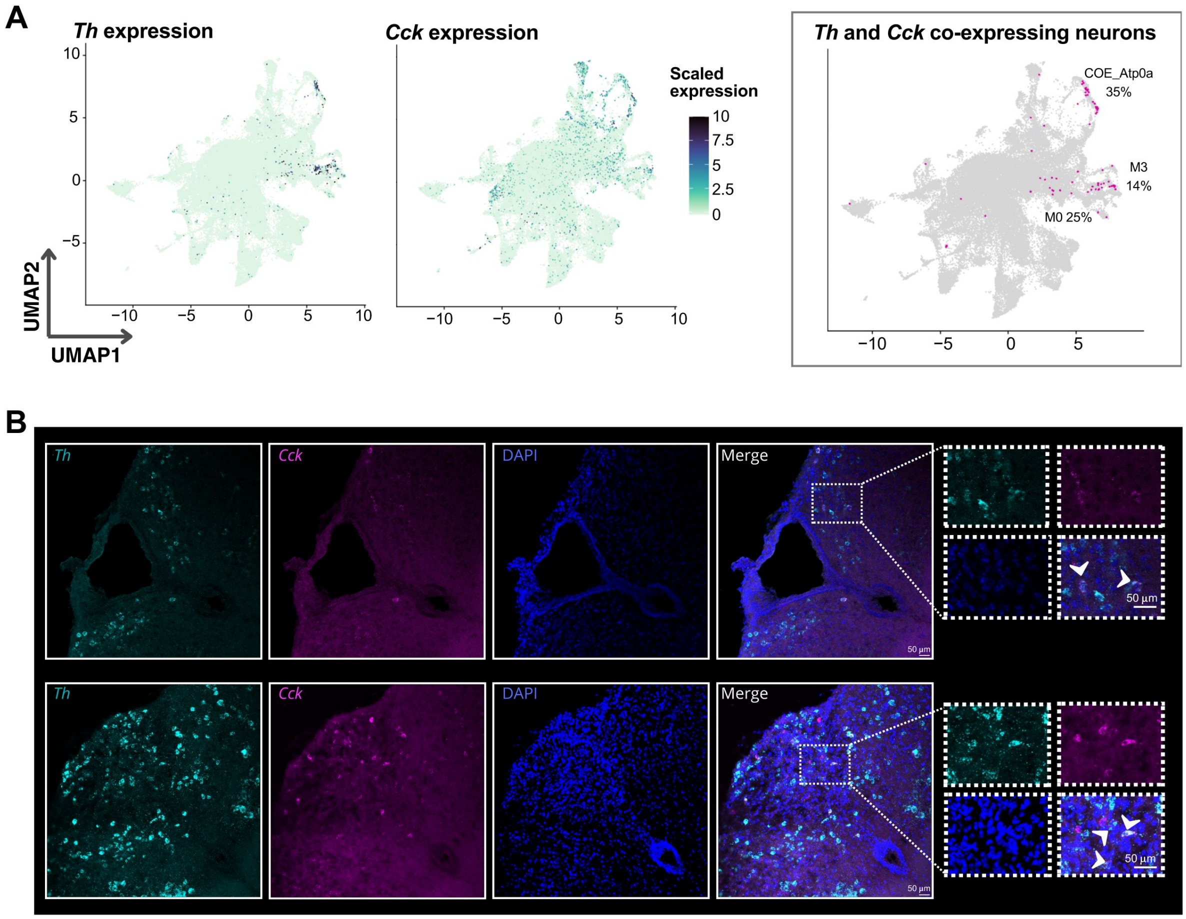

Figure 4

Th and *Cck* co-expression in the DVC

A. UMAP plot of neuronal *Th* and *Cck* scaled expression and cells with co-expression of both genes (neurons n=48,131; *Th*-expressing neurons n=764; *Cck*-expressing neurons n=3,821; *Th*/*Cck* co-expressing neurons n=80). On the right plot, cells highlighted in magenta are the co-expressing neurons. The three neuronal cell identities in which the majority of nuclear co-expression of *Th* and *Cck* mRNA is found, as well as the percentage of total co-expressing neurons, are shown. **B.** *In-situ* hybridization of *Cck* and *Th* mRNA in coronal mouse DVC sections corresponding to -7.2mm and -7.56mm relative to bregma showing overlap of *Th*/*Cck*. Some of the cells with overlapping signal are highlighted with an arrow in the enhanced merged image. DVC=dorsal vagal complex; UMAP=uniform manifold approximation and projection

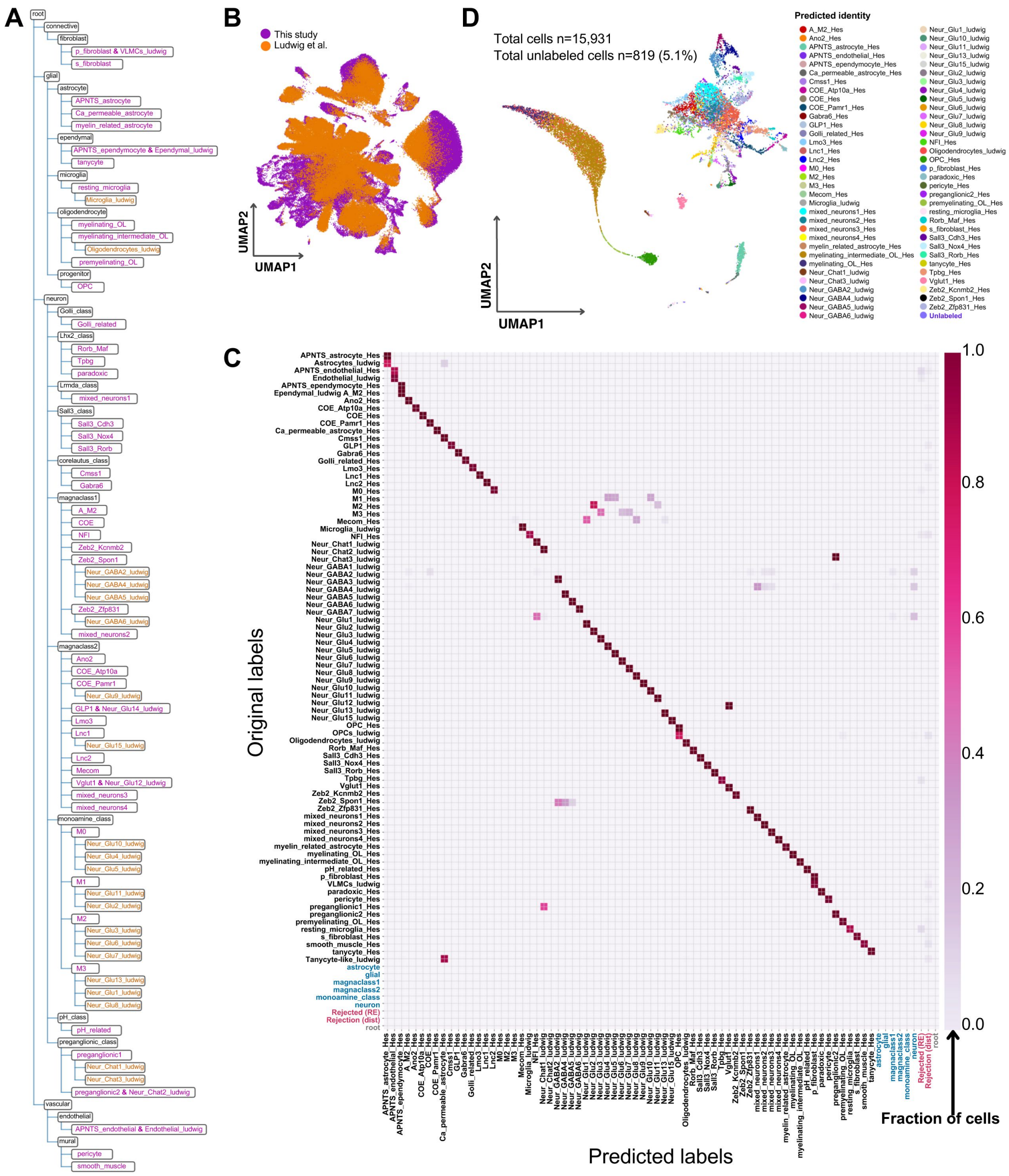

Figure 5

The murine DVC cell hierarchy

A. Dendrogram of the harmonized hierarchy which incorporates cells from Ludwig and our datasets. Orange cell identities are a fourth layer of cell identity resolution obtained as some of the Ludwig dataset identities are subgroups of our original cell identities at their highest resolution. Magenta labels represent layer 3 of cellular granularity. These two layers are considered high-resolution. **B.** UMAP plot of the integration between our and Ludwig datasets using treeArches (cells n=171,868). **C.** Pairwise heatmap showing the proportion of cells originally labeled in this study and in Ludwig dataset (y-axis), predicted to belong to each identity group (x-axis) by treeArches using our learned harmonized hierarchy. In blue are the labels considered non-specific for a high-resolution cell identity (n=2,324; 1.3%). In pink are the cell labels for rejected cells, therefore not assigned any identity by treeArches (n=3,108; 1.8%). **D**. UMAP plot of the Dowsett dataset labeled through treeArches using the learned harmonized hierarchy representation from our murine DVC atlas. The 'unlabeled' cells include those rejected by the algorithm and thus without an assigned identity, and those with an unspecific layer-3/4 label. For example, some were labeled 'neuron' or 'monoamine-class' but without a high-resolution cell identity.
UMAP=uniform manifold approximation and projection; APNTS=Area postrema and nucleus of the solitary tract; Lnc=long non-coding; OL=oligodendrocyte; OPC=oligodendrocyte precursor cell; Neur=neuronal; COE=collier/Olf1/EBF transcription factor

We confirmed the validity of our cell hierarchy using three methods. First, by using the learned model to predict the labels of the cells we used to develop them (i.e. this study and the Ludwig dataset), for which we obtained highly concordant cell labeling (**Fig.5C**). Secondly, we used the learned model to predict the labels of another publicly available dataset from Dowsett and collaborators (i.e. the 'Dowsett dataset')[18] (**Fig.5D**). Using this approach, we were able to successfully label 95% of these cells at high resolution (i.e. layer 3 or 4 of cell identity granularity); further validating our combined DVC labels. Finally, we manually mapped the marker genes from Ludwig[2] and Dowsett[18] neuronal clusters in our dataset and confirmed the sub-groups of neurons among our identities pointed out by treeArches (Fig.S11B).

3.4. An integrated rodent DVC cell hierarchy

To identify rat DVC cell clusters and subclusters and contrast these with the mouse DVC atlas, we generated a *de novo* ad libitum-fed rat dataset (cells n=12,167) (**Fig.6A**) and labeled it by mapping the top 5 gene expression markers of each of our murine layer-3 cell identities (Table S8). After manual curation at three granularities (**Fig.6B-C**; Table S9), we obtained 52 DVC rat identities at high cellular resolution (**Fig.6C**) including a small group of 'unspecific' cells (n=35) that we could not confidently label because its top gene expression markers have unknown functions and none of our known identities' genes clearly marked this population (**Fig.6B-C**; Table S8).

We established 10, layer-3 neuronal identities not previously found in the murine data (**Fig.6C**; Fig.S12A; Table S9). Notably, two of those did not belong to any of the existing murine neuronal classes (i.e. layer-2 of cellular granularity), therefore we created two new layer-2 classes (**Fig.6B**). One of these, the immunity-related neurons, have high *Csf1r* expression, as well as cystatin C and *Inpp5d* which are commonly associated with immunity-related processes in microglia and with immunomodulation in cancer[64, 65] (**Fig.7**; Table S10). The second neural class identified in rat but not mouse DVC were the 'Pdgfra neurons' (**Fig.7**). These neurons express *Pdgfra*, *Arhgap31* and *Itga9*, known markers for progenitor cells for which we named this cell class 'ortus-akin' (Latin: 'origin') because they are similar to cells giving origin to other cells[66] (**Fig.7A**; Table S10). Interestingly, this neuronal class not only expressed *Pdgfra*, but also other OPC signature genes (**Fig.7A**). Mapping the expression levels of metabolic-associated receptors revealed that a subset (~15%) of both immunity-akin and ortus-akin express detectable *Gfral* mRNA but this is likely an underrepresentation due to the challenges in detecting lowly expressed transcripts such as *Gfral*. Prolactin receptor and leptin receptor (**Fig.7B**) were also found to be expresses by these neurons, although they seem to have a varied neurotransmitter profile (Fig.S12B). To confirm the presence of *Pdgfra*-expressing neurons in the rat DVC, we performed immunostaining in the mouse and rat brainstem. Within the mouse, we found PDGFRA-immunoreactivity(IR) in the two morphologically distinct cells; matching OPC and blood vessel-associated fibroblast morphologies(**Fig.7C**). However, within the rat, we identified PDGFRA-IR in cells with three morphologies, one of which resembled neurons with an enlarged cell body and protruding cellular processes (**Fig.7C**). To determine if these PDGFRA-IR cells were neurons, we co-stained with the neural marker, HuC/D and noted several PDGFRA and HuC/D copositive cells (**Fig.7D**); confirming our sequencing data that a PDGFRA-expressing neural class exists in the rat DVC and localizing these cells to the area postrema.

Other differences found between mouse and rat DVC expression includes increased expression of the leptin receptor in neurons as well as cholecystokinin (Fig. S13).

To construct an integrated rodent DVC cell atlas, we incorporated the labeled rat dataset into our hierarchy using treeArches[39] (**Fig.8**). The rat data was projected onto the mouse datasets (**Fig.8A**) and the 42 high resolution rat cell identities were appended to the tree (**Fig.8B**; Fig.S13). The final hierarchy has 123 labels of which 99 are high resolution according to our cellular granularity (i.e. layers 3 to 5) (**Fig.8B**). In addition it has 20 layer-2 identities and 7 cell identities that are novel from the rat dataset (**Fig.8B**).

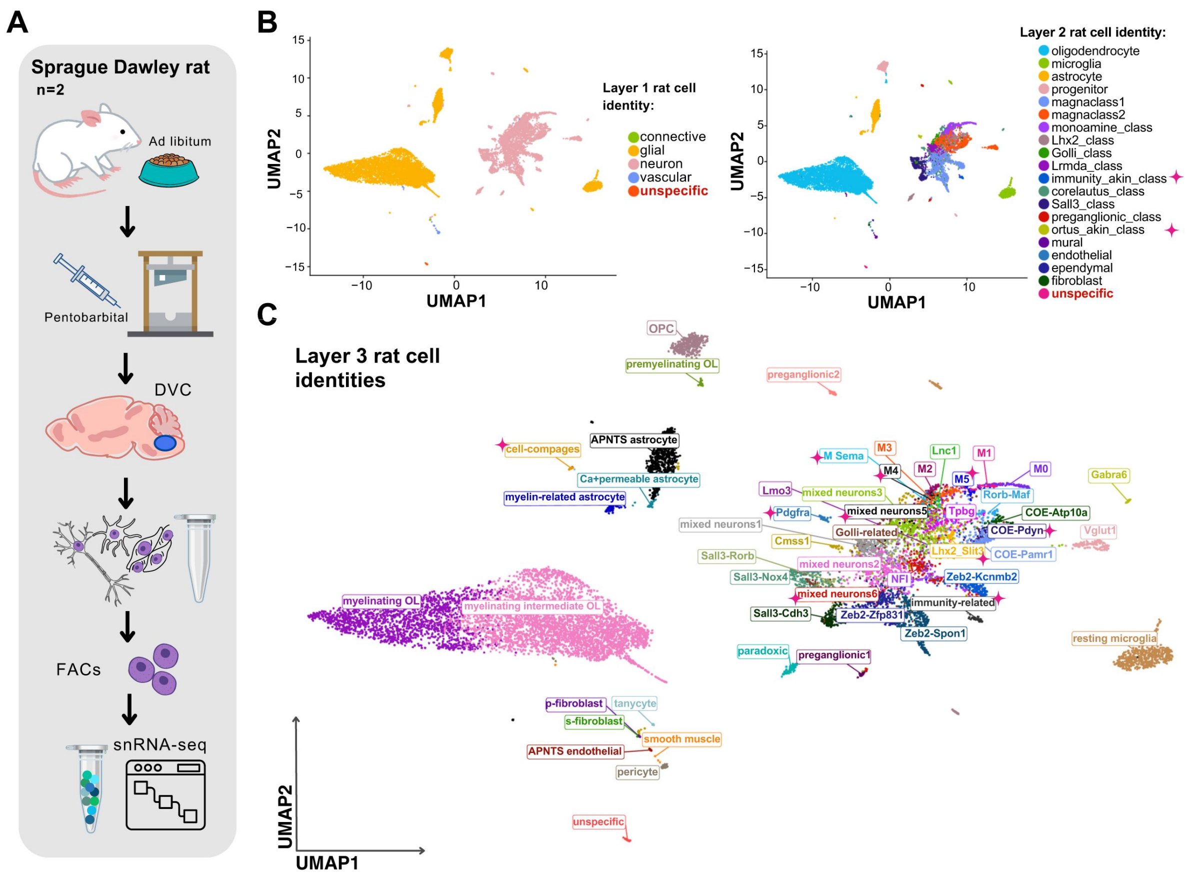

Figure 6

The snRNA-seq derived cell identities for the rat DVC

A. Schematic of the pipeline for snRNA-seq of the rat DVC. **B**. Labeled UMAP plots of the low resolution (i.e. layers 1 and 2 of granularity) and **C**. high resolution (i.e. layer 3) cell identities in our rat DVC dataset (cells n=12,167). We labeled four layer-1, nineteen layer-2 and fifty-two layer-3 cell identities. Those labels novel to this dataset and not present in the murine data are highlighted with a pink ♦ symbol. We found a small cluster (cells n=35) that we could not corroborate its identity at any layer of cellular granularity that we named 'unspecific'.

snRNA-seq=single-nuclei RNA-sequencing; DVC=dorsal vagal complex; FACs=fluorescence-activated cell sorting; UMAP=uniform manifold approximation and projection; APNTS=Area postrema and nucleus of the solitary tract; Lnc=long non-coding; OL=oligodendrocyte; OPC=cursor cell; Neur=neuronal; COE=collier/Olf1/EBF transcription factors

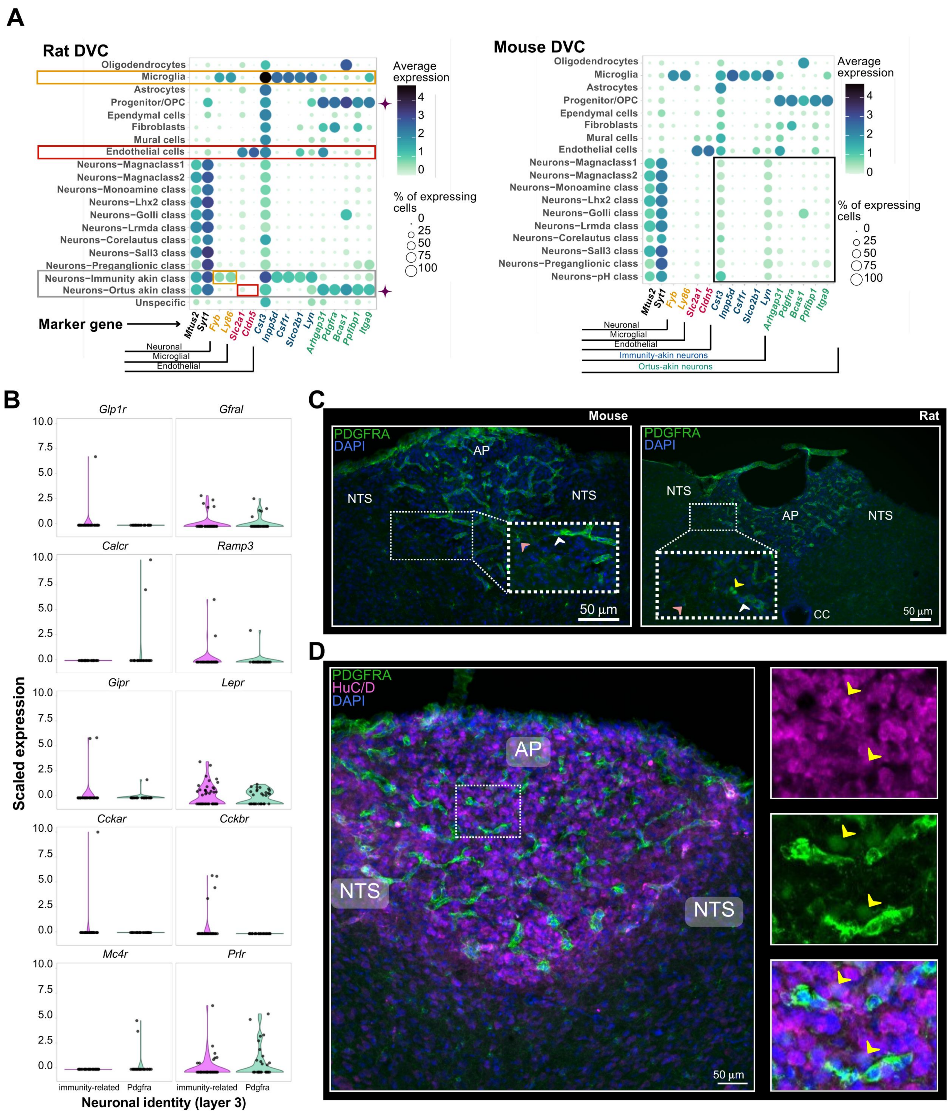

Figure 7

Two novel neuronal classes specific to the rat DVC

A. Balloon plots comparing the expression of the marker genes of the two novel neuronal classes found in rats (cells n=12,167): immunity-akin and the ortus-akin classes (framed in gray). Expression is shown across the layer-2 rat cell identities. Based on the expression of two DVC neuronal markers *Mtus2* and *Syt1*, we corroborated those cell classes contain neurons, and not microglial or vascular/endothelial cells (In orange and red, respectively). The immunity-akin class (n=48 neurons) shares high expression of *Cst3*, *Inpp5d*, *Csf1r*, *Slco2b1* and *Lyn* with microglial cells. Meanwhile, the ortus-akin class (n=34 neurons) shows higher overlap of gene expression markers (i.e. *Arhgap31*, *Pdgfra*, *Bcas1*, *Ppfibp1* and *Itga9*) with OPCs (highlighted with a ♦ symbol). Mouse cell identities (cells n=99,740) do not show these expression patterns (framed in black). Average expression was calculated using log-normalized counts. **B**. Violin plots of scaled expression of 10 genes (i.e. *Glp1r, Gfral, Calcr, Ramp3, Gipr, Lepr, Cckar, Cckbr,Mc4r* and *Prlr*) coding for metabolism-associated receptors in the Pdgfra and immunity-related rat neurons (Pdgfra neurons n=34; immunity-related neurons n=48). Pdgfra neurons are the only layer-3 identity within the ortus-akin class. An overlapping dotplot shows one dot per cell. **C**. Immunofluorescent detection of PDGFRA in mouse and rat DVC. Pink arrowheads point to cells with OPC morphology, white arrowheads point to cells with fibroblast morphology and the yellow arrowhead points to cells with neural morphology. **D.** Co-staining for PDGFRA and HuC/D in rat DVC with high magnification (right images) of area postrema PDGFRA-expressing cells . Yellow arrowheads indicate colocalization of HuC/D and PDGFRA.

DVC=dorsal vagal complex; OPC=oligodendrocyte precursor cell; AP=area postrema; NTS=nucleus of the solitary tract; PDGFRA= Platelet-derived growth factor receptor A; CC=central canal

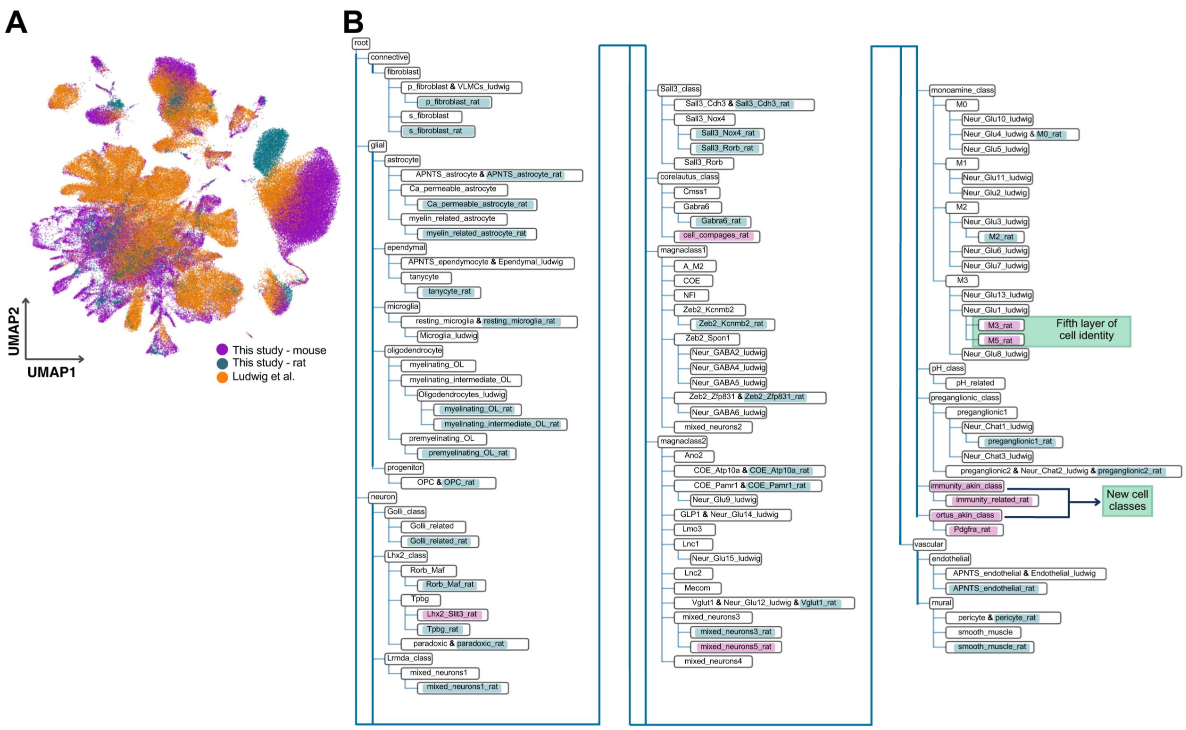

Figure 8

The rodent DVC cell hierarchy

A. UMAP plot of the integration between the murine hierarchy and our rat dataset using treeArches (cells n=184,035). **B.** Dendrogram of the harmonized hierarchy of our labeled rat dataset and the murine DVC hierarchy. We highlight the incorporated rat cell identities (blue and magenta). Those in magenta are the novel rat identities established by us. The immunity-akin and the ortus-akin classes were not found in the murine datasets. The M3 and M5 rat identities are subgroups of a Ludwig layer 4 cell identity, therefore yields a fifth layer within the monoamine neuronal class.
UMAP=uniform manifold approximation and projection; DVC=dorsal vagal complex; APNTS=Area postrema and nucleus of the solitary tract; Lnc=long non-coding; OL=oligodendrocyte; OPC=oligodendrocyte precursor cell; Neur=neuronal; COE=collier/Olf1/EBF transcription factors

3.5. *Hcrt* and *Agrp* expression in the DVC

Since *Agrp*-expressing cells have been recently described within the DVC[67], we mapped this and other neuropeptide genes in DVC neurons. Surprisingly, we found *Hcrt* and *Agrp* mRNA in neurons in our dataset (Fig.S14A). We then performed *in-situ* hybridizations for these transcripts in both the mouse hypothalamus and DVC. While both *Hcrt* and *Agrp* were readily detectable in the hypothalamus, we failed to observe signal within the DVC (Fig.S14B). Since we also found expression of *Hcrt* and *Agrp* in neurons of the Dowsett and Ludwig datasets[2, 18] (Fig.S14C), we then mapped the sequenced reads from these transcripts and confirmed that most of them mapped to intronic and non-coding regions of these genes (Fig.S14D). Furthermore, our rat DVC dataset was pre-processed excluding introns from the reads mapping, and the *Hcrt* and *Agrp* expression levels observed is minimal (Fig.S14E), which is in agreement with the non-coding location of the reads in mouse DVC. These data explain absence of signal from our *in-situ* hybridizations and suggests DVC cells do not produce HCRT or AGRP neuropeptides.

3.6. Meal-related transcriptional programs in the murine DVC

As the DVC acts as a primary node for meal-related signals[68], we wanted to define which of our cell identities responded to meal consumption and thus performed differential gene expression analysis between DVC cell types from mice that were euthanized under fasting conditions, with ad libitum access to food or two hours post-refeeding following an overnight fast (**Fig.1A**; **Fig.9**). This analysis revealed widespread transcriptional changes across nearly every neural cell identity, the exceptions being corelautus, Golli class and pH classes (**Fig.9A**). Additionally, the size of the transcriptional changes were relatively small, with approximately 90% of the differentially expressed genes having increases or decreases between 0.5 and 1.0 log2 fold-change, with only ~10% of the gene changes being ≥1 log2 fold-change between conditions (**Fig.9A**). Of the responding neurons, magnaclasses 1 and 2, which represent large populations of predominately inhibitory and excitatory neurons (Fig.S2), showed the greatest number of differentially expressed genes following refeeding for both comparisons refed versus fasting, and refed versus ad libitum food access (**Fig.9A**). To understand how refeeding transcriptionally alters inhibitory compared with excitatory DVC neurons, we compared differentially expressed genes between these two classes across conditions. Intriguingly, despite being in transcriptionally distinct classes, we found meal consumption altered many of the same genes (**Fig.9B-D**), suggesting these cells may be receiving similar postprandial signals which are propagated via these cells in unique manners. This is supported by the similar receptor expression profile of magnaclass 1 and 2 (Fig.S15). One notable exception is *Cckbr* which is upregulated in magnaclass 2 and downregulated in magnaclass 1 in refeeding when compared with ad libitum fed mice(Table S12). We also note that oligodendrocytes contained the highest number of differentially expressed genes in refeeding (**Fig.9A**). While the roles of oligodendrocytes in refeeding responses within the DVC have yet to be determined and the biologic significance of these changes are presently unclear, DVC oligodendrocytes also have robust transcriptional responses in response to fasting[18]; suggesting these cells are sensitive to both positive and negative energy balance associated-cues and may have important roles in regulating energy homeostasis.

4. Discussion

The recent success of DVC-based therapies[13, 69] has moved DVC biology to the forefront of weight control therapies and there is now clear clinical impetus to fully define DVC cell types and their functions. Previous studies detailing DVC cell types describe a high degree of heterogeneity[2, 18, 19], however, these studies lacked an in-depth analysis of

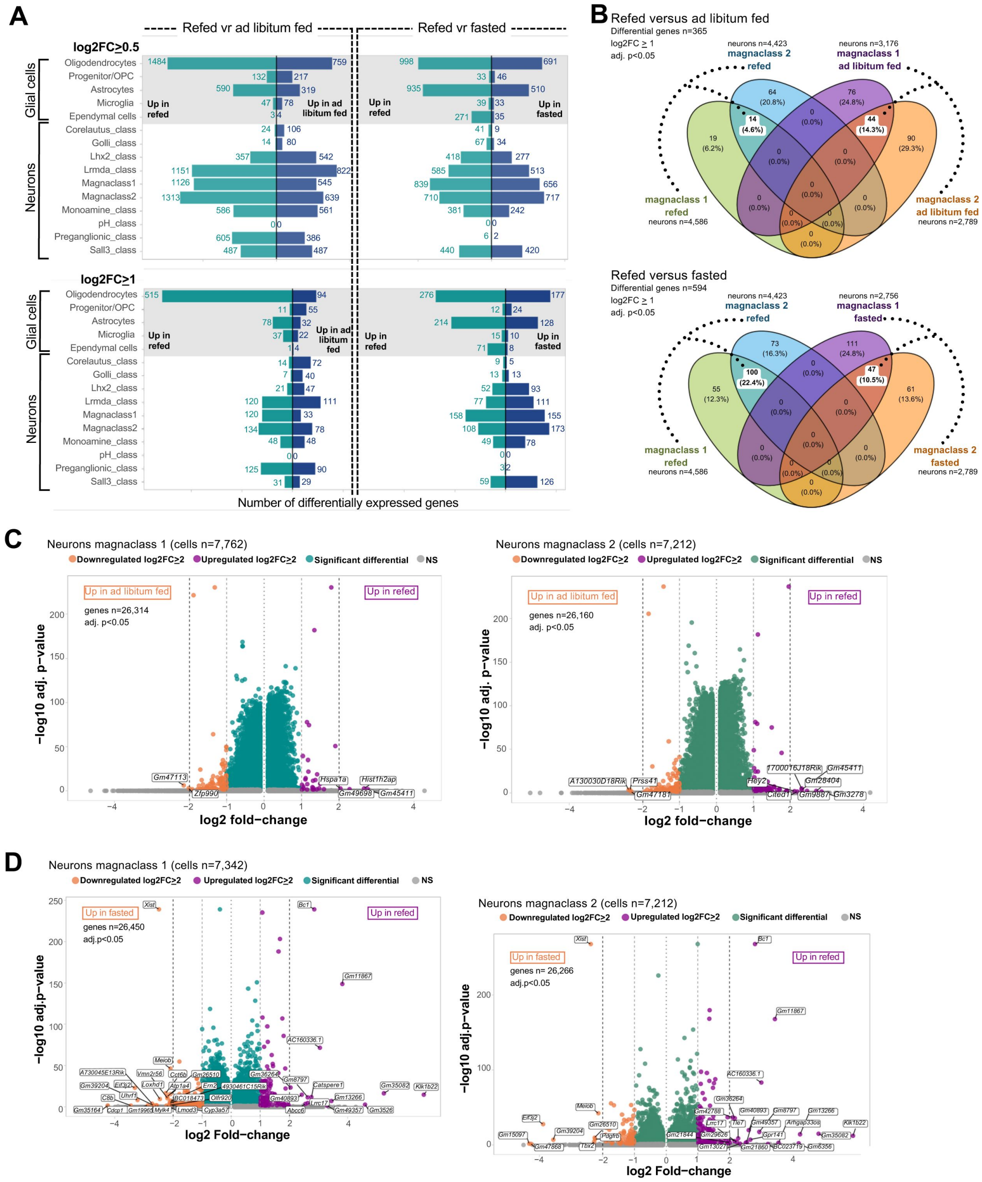

Figure 9

Meal-related transcriptional changes in the mouse DVC

A. Horizontal barplots of the number of differential genes between refed and ad libitum fed, and refed and fasted mice in glial cells and neurons per layer-2 cell identity (MAST algorithm on log-normalized counts; adj. p<0.05; refed neurons n=14,396; refed glial cells n=18,019; ad libitum fed neurons n= 9,885; ad libitum fed glial cells n=4,610; fasted neurons n=8,829; fasted glial cells n=12,496). **B.** Venn Diagrams of the differentially expressed genes in magnaclass 1 and magnaclass 2 neurons between refed and ad libitum fed, and between refed and fasted treatments (MAST algorithm on log-normalized counts). The number of overlapping log2FC≥1 upregulated genes between the two magnaclasses per treatment are highlighted. The treatment-induced changes in only one cell class are shown as non-overlapping. The percentage is based on the total differential genes surveyed per comparison. **C.** Volcano plots of the differential genes in neurons belonging to the magnaclass 1 and the magnaclass 2 between refed and ad libitum fed, and **D.** refed and fasted treatments (MAST algorithm on log-normalized counts; adj. p<0.05). Each point represents one gene. Only genes upregulated or downregulated with a log2FC≥2 are labeled. Since we considered a minimum log fold-change of 0.1 between treatments per cell group, those genes with very low variance (i.e. log fold-change<0.1) were excluded from the differential expression analysis, therefore a variable number of genes are shown per comparison per identity, in the volcano plots.

DVC=dorsal vagal complex; vr=versus; OPC=oligodendrocyte precursor cell; FC=fold-change; adj.=adjusted; NS=non-significant neurotransmitter profiles of neurons, any cross-species evaluation of DVC cell types, or an analysis of how meal consumption transcriptionally regulates these cells. We therefore developed *de novo* mouse and rat DVC cell atlases to address these points.

To more fully describe the cellular complexity within the DVC, we combined a pipeline of cell label transfer using brain existing published datasets and thorough curation of cell identities, which resulted in a 4-layer mouse cell hierarchy and a 5-layer rodent cell hierarchy of the DVC. Within this hierarchy we found several unique features of DVC glial cells. First, we found Ca^{+}-permeable astrocytes in the DVC in which *Gfap* is almost exclusively expressed and they coexpress *Kcnj3* (GIRK1 potassium channel). These cells largely lack *Gria2*, consistent with findings in Bergmann glial cells in which colocalization of *Gria2* and GFAP was found to alter neuron-glial interactions[70]. The heterogeneity of astrocytes across, and even within, spatially distinct CNS sites is becoming increasingly apparent[71] and our finding of DVC-specific astrocyte transcriptional profiles further supports this. When we compared Ca^{+}-permeable DVC astrocytes to hypothalamic astrocytes[43], we found the same expression pattern with minimal overlap between *Gria2* and *Gfap*. However, DVC astrocytes seem to have some unique properties when compared with hypothalamic, cortical and hippocampal astrocytes. Principally, DVC astrocytes express the GIRK1 channel, likely rendering them less excitable[72]. As astrocytes have well established roles in maintaining the blood brain barrier function[73], and the pattern of GFAP in the DVC is largely restricted to the boundary separating the AP and NTS, it is possible these DVC-specific astrocytes have specialized functions in regulating blood brain barrier permeability.

Additionally, while we identified hypothalamic fibroblasts which express high *Aldh1a1* similar to the DVC p-fibroblasts, we failed to find a group equivalent to s-fibroblasts. Other cell populations seem to be consistent to what has been found in other brain sites, like the intermediate oligodendrocytes that are found enriched in the hypothalamus and the corpus callosum in mice[55]. In summary, these findings show that while certain DVC glial cells share identities with glial cells from other CNS sites, there are also DVC-specific glial cells.

Within the mouse neural classes, we found overlap between *Cck* and *Th*-expressing cells. As these cells have previously been described as non-overlapping, this was unexpected[17]. Our *in-situ* hybridization confirmed the presence of *Cck*/*Th* positive cells within the NTS and AP. We believe the discrepancy may be due to detection methods as these cells were initially defined as non-overlapping using immunofluorescent detection of TH protein and a *Cck* genetic reporter[17]. Functional studies of *Cck*- and *Th*-expressing neurons within the NTS demonstrate these cells have somewhat distinct roles; while both reduce appetite, *Cck* neurons are strongly aversive whereas *Th*-labelled neurons are non-aversive[74]. While no functional characterization of *Cck* and *Th* neurons of the AP has been performed, we posit that as some *Th*-expressing AP neurons also express *Gfral* and *Casr*; both having potently aversive ligands (e.g. GDF15, deoxynivalenol)[16, 75]; they are likely to mediate aversive anorexia. The role of dual *Cck/Th* positive neurons remains to be determined.

While AGRP expression has long been considered to be hypothalamic-exclusive, a recent report describes *Agrp*-expressing cells within the mouse DVC[67], i n part from the detection of Agrp mRNA in scRNA-Seq data. Similarly, we also detected transcripts for both *Agrp* and *Hcrt* in our snRNA-Seq data but failed to detect any transcripts within the DVC by *in-situ* hybridizations. One possible explanation for the discrepancy between the snRNA-Seq data and our *in-situ* hybridizations data is that the snRNA-Seq mRNA reads for*Agrp* and *Hcrt* gene overwhelmingly mapped to non-coding regions, of the respective genes. Combined, these data suggests to us that neither AGRP or HCRT neuropeptides are produced by DVC cells in rodents.

We show the presence of neurons with seemingly mixed programs in the DVC, which we called ‘mixed neurons’. Similar groups have been described in the hypothalamus[26] suggesting that some neurons may have hybrid-functionalities. These mixed neurons share many transcriptomic markers expression with many other cell identities, and may require the use of other technologies to fully identify them anatomically in the brain.

To perform a cross-species analysis, we generated a rat-based DVC atlas and compared it with the mouse DVC. Our rat dataset was smaller than the mouse data, it is possible some of the of the new neuronal cell identities (e.g. M_sema, M4) may contain cells belonging to multiple cell types, as treeArches did not appended these identities to the tree. Despite this, we corroborated two new cell types in the rat DVC that are not present in mice the *Pdgfa*-expressing (ortus-akin class) and the immunity-related neurons. *Pdgfra* is a well described marker gene for OPCs and fibroblasts within the central nervous system[66] but, to our knowledge, has not been associated with neurons. Notably, although the PDGFRA immuno-signal in HuC/D+ neurons is clearly discernible, it is less intense than in OPCs and fibroblasts, in agreement with expression levels from our snRNA-Seq data. Since it was previously demonstrated that OPCs can give rise to neurons in the adult brain[76] and there is evidence of neurogenesis in the AP following amylin administration in rats[77], it is possible that these PDGFRA+/HuCD+ cells in the AP represent a transient stage of PDGFRA+ OPCs differentiating into neurons.

Regardless of their maturity, we speculate that the Pdgfra neurons (ortus-akin class) have roles in appetite regulation as they express leptin receptor and *Gfral*. Interestingly, while multiple studies have failed to detect *Lepr* via Cre-based reporters[62, 78] or leptin-induced phosphorylated STAT3 in the AP of mice[79], *Lepr* and leptin-induced phosphorylated STAT3 are detectable within the rat AP[80–82]. We believe one explanation for this species discrepancy in *Lepr* expression is the absence of the ortus class neurons in mice. Future studies are needed to characterize these cells and determine whether their presence is found in other mammals including humans.

DVC-targeting obesity therapeutics, including amylin and GLP1 mimetics, produce clinically relevant weight loss but are associated with multiple gastrointestinal adverse events and there is a need to develop new obesity therapies that can reduce appetite without unwanted gastrointestinal side effects[83]. Indeed, the DVC cells that control appetite suppression and nausea are separable[11, 74], making targeting such cells an attractive target for pharmacotherapy. As meal consumption is generally not associated with gastrointestinal distress, we sought to identify refeeding responsive cells within the mouse DVC. From this analysis, we found nearly all neural cell identities are transcriptionally altered by meal consumption. We also found the magnitude of transcriptional changes was relatively small, regardless of the cell class examined. While relatively small, the scale of the transcriptional changes in response to refeeding are similar to many of the gene expression changes observed in *Agrp*-neurons following 16 hours food deprivation[43]. Regardless of whether this is a technical limitation of snRNA-Seq methods or accurately reflects the scale of transcriptional changes in the DVC, utilizing transcriptional responses to refeeding to identify neural classes as targets for pharmacotherapy is unlikely to yield specific and targetable pathways.

In general, one limitation in our study is that it is based on nuclear RNA which may reflect a fraction of biological changes. Using nuclear isolation facilitates generating to have single-neuron data[84] since obtaining viable whole neurons (including all their processes) is very challenging. Previous studies on the relationship between nuclear and cellular RNA content in brain show that nuclear sequencing reads may map to intronic regions of genes in higher proportion[85]. Although this does not appear to impact gene expression analysis for neurons in the current pipelines like the ones we used for our mouse data[85], it is possible that some biological features such as microglial activation state, are not fully recovered using nuclear RNA[86]. Although we performed sub-clustering in the mouse microgia and assessment of the resulting marker genes for each group, we did not find evidence of activation possibly due to this matter. Separately, for meal consumption, we were unable to analyze responses in individual mice as pooling DVCs from individual mice was the most amenable option to maximize nuclei for in our pipeline. As we pooled DVCs from both sexes, this limited our ability to detect sexual dimorphic cell class proportions and gene expression levels.

5. Conclusion

As the amount of snRNA-Seq data performed on the DVC continues to grow, there is a need for a common, scalable label set that can be applied to new studies and integrate new sequencing data. To develop such a label set, we started with a *de novo* DVC atlas, we generated three layers of cellular granularity for both neural and non-neuronal cell types. Then, using the treeArches pipeline[39] we harmonized our snRNA-Seq labels with the Ludwig dataset[2] and increased our cellular granularity from three to four layers of resolution. Manually curating and harmonizing our dataset enhanced the ability of our cell atlas to capture the complexity of the DVC neurons since not all groups seem to contain easily delineable subgroups of cells. For example, eleven of the labels incorporated into our cell hierarchy in this pipeline belong to the monoaminergic neuronal class, which seem to encompass multiple subgroups of specialized cells. We further constructed the rodent DVC cell atlas which resulted in a fifth layer of cell identity to the monoaminergic M3 neurons. Although the number of identities increases as layers are more specific, this gives this unified atlas greater adaptability to different research needs. Additionally, our labels can be accurately harmonized with new databases regardless of size and new DVC snRNA-Seq data can be added to the tree to generate new branches. Thus, our unified rodent DVC atlas is highly amenable to future research elucidating new biologic and therapeutic insights into DVC cell types.

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Additional information

7. Author contributions

Conceptualization: CH and PS

Methodology: CH, AB, LM, MK and PS

Sample collection: AB and PS

Software: CH, LM and HJM

Data analysis: CH, LM, MK and PS

Data curation: CH and HJM

Validation of findings: CH, FS, HJM and PS

Imaging: CH and PS

Visualization: CH, MK and PS

Supervision: CH and PS

Writing—original draft: CH and PS —review and editing: CH, AB, LM, FS, HJM, MK and PS

Funding acquisition: MK and PS

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10. Data statement

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. FASTQ files have been deposited in the NCBI Sequence Read Archive (Accession: PRJNA1161292). In addition, our murine and rat datasets have been made available for visualization and analysis through the Broad Institute Single Cell Portal (*https://singlecell.broadinstitute.org/single_cell/study/SCP2773*). The labels from our cell atlas can be transferred to new datasets through treeArches using the available hosted Jupyter Notebook in Google Colab for the mouse data (*https://colab.research.google.com/drive/1v983gcQxIwBN-4lGx18JXPweZoKqMbwX#scrollTo=SJTZppZWa-63*), and for the rodent data (*https://colab.research.google.com/drive/10tAmfRosMscuBak6wHX80TgdEBsHeWQB#scrollTo=941ab5c1-8910-4e67-a632-4d18601d45b4*). Original code used for constructing the murine and rodent cell hierarchies and predicting labels using treeArches are available in the GitHub repository *https://github.com/LabSabatini/DVC_cell_atlas*.

Abbreviations

- 3v: third ventricle
- adj.: adjusted
- AMPAr: alpha-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor
- AP: area postrema
- APNTS: Area postrema and nucleus of the solitary tract
- ARH: arcuate nucleus of the hypothalamus
- BAM: binary alignment map
- CC: central canal
- COE: collier/Olf1/EBF transcription factor
- diH_2O: distilled water
- DMV: dorsal motor nucleus of the vagus
- DVC: dorsal vagal complex
- FACS: fluorescence-activated cell sorting
- FC: fold-change
- GABA: gamma-aminobutyric acid
- GDF15: growth differentiation factor 15
- GIRK1: G protein-coupled inwardly rectifying potassium channel 1
- GLP1R: glucagon-like peptide 1 receptor
- GFRAL: GDNF family receptor alpha-like
- IR: immunoreactivity

- LH: lateral hypothalamus
- Lnc: long non-coding
- ME: median eminence
- Neur: neuronal
- NG: *Ng2* glial cell
- NS: non-significant
- NTS: nucleus of the solitary tract
- OL: oligodendrocyte
- OPC: oligodendrocyte precursor cell
- PC: principal component
- PDGFRA: Platelet-derived growth factor receptor A
- Res.: resolution
- RNA-seq: RNA sequencing
- smc: smooth muscle cells
- snRNA-seq: single-nuclei RNA sequencing
- UMAP: uniform manifold approximation and projection
- UMI: unique molecular identifiers
- v: version
- VGLUT2: vesicular glutamate transporter-2
- vr: versus

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Additional files

Supplemental Figures

References

[1] Gautron L., Elmquist J.K., Williams K.W. (2015) **Neural control of energy balance: translating circuits to therapies** *Cell* **161**:133–45 https://doi.org/10.1016/j.cell.2015.02.023 | Google Scholar

[2] Ludwig M.Q., Cheng W., Gordian D., Lee J., Paulsen S.J., Hansen S.N., et al. (2021) **A genetic map of the mouse dorsal vagal complex and its role in obesity** *Nat Metab* **3**:530–45 https://doi.org/10.1038/s42255-021-00363-1 | Google Scholar

[3] Bai L., Mesgarzadeh S., Ramesh K.S., Huey E.L., Liu Y., Gray L.A., et al. (2019) **Genetic Identification of Vagal Sensory Neurons That Control Feeding** *Cell* **179**:1129–43 https://doi.org/10.1016/j.cell.2019.10.031 | Google Scholar

[4] Han W., Tellez L.A., Perkins M.H., Perez I.O., Qu T., Ferreira J., et al. (2018) **A Neural Circuit for Gut-Induced Reward** *Cell* **175**:665–78 https://doi.org/10.1016/j.cell.2018.08.049 | Google Scholar

[5] Williams E.K., Chang R.B., Strochlic D.E., Umans B.D., Lowell B.B., Liberles S.D. (2016) **Sensory Neurons that Detect Stretch and Nutrients in the Digestive System** *Cell* **166**:209–21 https://doi.org/10.1016/j.cell.2016.05.011 | Google Scholar

[6] Schwartz M.W. (2006) **Central nervous system regulation of food intake** *Obesity* **14**:1–8 https://doi.org/10.1038/oby.2006.275 | Google Scholar

[7] Cheng W., Ndoka E., Maung J.N., Pan W., Rupp A.C., Rhodes C.J., et al. (2021) **NTS Prlh overcomes orexigenic stimuli and ameliorates dietary and genetic forms of obesity** *Nat Commun* **12**:5175 https://doi.org/10.1038/s41467-021-25525-3 | Google Scholar

[8] Yanovski S.Z., Yanovski J.A. (2014) **Long-term drug treatment for obesity: a systematic and clinical review** *JAMA* **311**:74–86 https://doi.org/10.1001/jama.2013.281361 | Google Scholar

[9] Borner T., Liberini C.G., Lutz T.A., Riediger T. (2018) **Brainstem GLP-1 signalling contributes to cancer anorexia-cachexia syndrome in the rat** *Neuropharmacology* **131**:282–90 https://doi.org/10.1016/j.neuropharm.2017.12.024 | Google Scholar

[10] Advani S.M., Advani P.G., VonVille H.M., Jafri S.H. (2018) **Pharmacological management of cachexia in adult cancer patients: a systematic review of clinical trials** *BMC Cancer* **18**:1174 https://doi.org/10.1186/s12885-018-5080-4 | Google Scholar

[11] Huang K.P., Acosta A.A., Ghidewon M.Y., McKnight A.D., Almeida M.S., Nyema N.T., et al. (2024) **Dissociable hindbrain GLP1R circuits for satiety and aversion** *Nature* **632**:585–93 https://doi.org/10.1038/s41586-024-07685-6 | Google Scholar

[12] Cheng W., Gordian D., Ludwig M.Q., Pers T.H., Seeley R.J., Myers M.G. (2022) **Hindbrain circuits in the control of eating behaviour and energy balance** *Nat Metab* **4**:826–35 https://doi.org/10.1038/s42255-022-00606-9 | Google Scholar

[13] Veniant M.M., Lu S.C., Atangan L., Komorowski R., Stanislaus S., Cheng Y., et al. (2024) **A GIPR antagonist conjugated to GLP-1 analogues promotes weight loss with improved**

metabolic parameters in preclinical and phase 1 settings *Nat Metab* **6**:290–303 https://doi.org/10.1038/s42255-023-00966-w | Google Scholar

[14] Adams J.M., Pei H., Sandoval D.A., Seeley R.J., Chang R.B., Liberles S.D., et al. (2018) **Liraglutide Modulates Appetite and Body Weight Through Glucagon-Like Peptide 1 Receptor-Expressing Glutamatergic Neurons** *Diabetes* **67**:1538–48 https://doi.org/10.2337/db17-1385 | Google Scholar

[15] Medic I., Gulati S., Lenz H.-J., Mahalingam D., Thomas J., Luo J., et al. (2022) **Initial results of a cohort of advanced prostate cancer patients in a phase Ia study of NGM120, a first-in-class anti-GDNF family receptor alpha like (GFRAL) antibody** *Annals of Oncology* **33**:S1186 https://doi.org/10.1016/j.annonc.2022.07.1888 | Google Scholar

[16] Yang L., Chang C.C., Sun Z., Madsen D., Zhu H., Padkjaer S.B., et al. (2017) **GFRAL is the receptor for GDF15 and is required for the anti-obesity effects of the ligand** *Nat Med* **23**:1158–66 https://doi.org/10.1038/nm.4394 | Google Scholar

[17] Roman C.W., Derkach V.A., Palmiter R.D. (2016) **Genetically and functionally defined NTS to PBN brain circuits mediating anorexia** *Nat Commun* **7**:11905 https://doi.org/10.1038/ncomms11905 | Google Scholar

[18] Dowsett G.K.C., Lam B.Y.H., Tadross J.A., Cimino I., Rimmington D., Coll A.P., et al. (2021) **A survey of the mouse hindbrain in the fed and fasted states using single-nucleus RNA sequencing** *Mol Metab* **53**:101240 https://doi.org/10.1016/j.molmet.2021.101240 | Google Scholar

[19] Zhang C., Kaye J.A., Cai Z., Wang Y., Prescott S.L., Liberles S.D. (2021) **Area Postrema Cell Types that Mediate Nausea-Associated Behaviors** *Neuron* **109**:461–72 https://doi.org/10.1016/j.neuron.2020.11.010 | Google Scholar

[20] Fortin S.M., Lipsky R.K., Lhamo R., Chen J., Kim E., Borner T., et al. (2020) **GABA neurons in the nucleus tractus solitarius express GLP-1 receptors and mediate anorectic effects of liraglutide in rats** *Sci Transl Med* **12** https://doi.org/10.1126/scitranslmed.aay8071 | Google Scholar

[21] Zheng G.X., Terry J.M., Belgrader P., Ryvkin P., Bent Z.W., Wilson R., et al. (2017) **Massively parallel digital transcriptional profiling of single cells** *Nat Commun* **8**:14049 https://doi.org/10.1038/ncomms14049 | Google Scholar

[22] R.C. Team (2023) **R: A Language and Environment for Statistical Computing** Vienna, Austria: R Foundation for Statistical Computing

[23] Hao Y., Stuart T., Kowalski M.H., Choudhary S., Hoffman P., Hartman A., et al. (2024) **Dictionary learning for integrative, multimodal and scalable single-cell analysis** *Nat Biotechnol* **42**:293–304 https://doi.org/10.1038/s41587-023-01767-y | Google Scholar

[24] McGinnis C.S., Murrow L.M., Gartner Z.J. (2019) **DoubletFinder: Doublet Detection in Single-Cell RNA Sequencing Data Using Artificial Nearest Neighbors** *Cell Syst* **8**:329–37 https://doi.org/10.1016/j.cels.2019.03.003 | Google Scholar

[25] Korsunsky I., Millard N., Fan J., Slowikowski K., Zhang F., Wei K., et al. (2019) **Fast, sensitive and accurate integration of single-cell data with Harmony** *Nat Methods* **16**:1289–96 https://doi.org/10.1038/s41592-019-0619-0 | Google Scholar

[26] Campbell J.N., Macosko E.Z., Fenselau H., Pers T.H., Lyubetskaya A., Tenen D., et al. (2017) **A molecular census of arcuate hypothalamus and median eminence cell types** *Nat Neurosci* **20**:484–96 https://doi.org/10.1038/nn.4495 | Google Scholar

[27] Chen R., Wu X., Jiang L., Zhang Y. (2017) **Single-Cell RNA-Seq Reveals Hypothalamic Cell Diversity** *Cell Rep* **18**:3227–41 https://doi.org/10.1016/j.celrep.2017.03.004 | Google Scholar

[28] Romanov R.A., Zeisel A., Bakker J., Girach F., Hellysaz A., Tomer R., et al. (2017) **Molecular interrogation of hypothalamic organization reveals distinct dopamine neuronal subtypes** *Nat Neurosci* **20**:176–88 https://doi.org/10.1038/nn.4462 | Google Scholar

[29] Tasic B., Menon V., Nguyen T.N., Kim T.K., Jarsky T., Yao Z., et al. (2016) **Adult mouse cortical cell taxonomy revealed by single cell transcriptomics** *Nat Neurosci* **19**:335–46 https://doi.org/10.1038/nn.4216 | Google Scholar

[30] Aran D., Looney A.P., Liu L., Wu E., Fong V., Hsu A., et al. (2019) **Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage** *Nat Immunol* **20**:163–72 https://doi.org/10.1038/s41590-018-0276-y | Google Scholar

[31] Risso D., Cole M. (2014) **scRNAseq: Collection of Public Single-Cell RNA-Seq Datasets** *Bioconductor* 2.16.0

[32] Finak G., McDavid A., Yajima M., Deng J., Gersuk V., Shalek A.K., et al. (2015) **MAST: a flexible statistical framework for assessing transcriptional changes and characterizing heterogeneity in single-cell RNA sequencing data** *Genome Biol* **16**:278 https://doi.org/10.1186/s13059-015-0844-5 | Google Scholar

[33] Cannoodt R., Saelens W., Sichien D., Tavernier S., Janssens S., Guilliams M., et al. (2016) **SCORPIUS improves trajectory inference and identifies novel modules in dendritic cell development** *bioRxiv* https://doi.org/10.1101/079509 | Google Scholar

[34] Ushey K., Allaire J., Tang Y. (2024) **reticulate: Interface to 'Python'** *CRAN* 1.35.0

[35] Cannoodt R. (2023) **anndata: 'anndata' for R** *CRAN* 0.7.5.6

[36] Van Rossum G., Drake F.L. (2009) **Python 3 Reference Manual** Scotts Valley, CA: CreateSpace Google Scholar

[37] Sathyamurthy A., Johnson K.R., Matson K.J.E., Dobrott C.I., Li L., Ryba A.R., et al. (2018) **Massively Parallel Single Nucleus Transcriptional Profiling Defines Spinal Cord Neurons and Their Activity during Behavior** *Cell Rep* **22**:2216–25 https://doi.org/10.1016/j.celrep.2018.02.003 | Google Scholar

[38] Batiuk M.Y., Martirosyan A., Wahis J., de Vin F., Marneffe C., Kusserow C., et al. (2020) **Identification of region-specific astrocyte subtypes at single cell resolution** *Nat Commun* **11**:1220 https://doi.org/10.1038/s41467-019-14198-8 | Google Scholar

[39] Michielsen L., Lotfollahi M., Strobl D., Sikkema L., Reinders M.J.T., Theis F.J., et al. (2023) **Single-cell reference mapping to construct and extend cell-type hierarchies** *NAR Genom Bioinform* **5** https://doi.org/10.1093/nargab/lqad070 | Google Scholar

[40] Gayoso A., Lopez R., Xing G., Boyeau P., Valiollah Pour Amiri V., Hong J., et al. (2022) **A Python library for probabilistic analysis of single-cell omics data** *Nat Biotechnol* **40**:163–6 https://doi.org/10.1038/s41587-021-01206-w | Google Scholar

[41] Michielsen L., Reinders M.J.T., Mahfouz A. (2021) **Hierarchical progressive learning of cell identities in single-cell data** *Nat Commun* **12**:2799 https://doi.org/10.1038/s41467-021-23196-8 | Google Scholar

[42] Hunter J.D. (2007) **Matplotlib: A 2D graphics environment** *Computing in Science & Engineering* **3**:90–5 https://doi.org/10.1109/MCSE.2007.55 | Google Scholar

[43] Steuernagel L., Lam B.Y.H., Klemm P., Dowsett G.K.C., Bauder C.A., Tadross J.A., et al. (2022) **HypoMap-a unified single-cell gene expression atlas of the murine hypothalamus** *Nat Metab* **4**:1402–19 https://doi.org/10.1038/s42255-022-00657-y | Google Scholar

[44] Jessa S., Blanchet-Cohen A., Krug B., Vladoiu M., Coutelier M., Faury D., et al. (2019) **Stalled developmental programs at the root of pediatric brain tumors** *Nat Genet* **51**:1702–13 https://doi.org/10.1038/s41588-019-0531-7 | Google Scholar

[45] Vaaga C.E., Borisovska M., Westbrook G.L. (2014) **Dual-transmitter neurons: functional implications of co-release and co-transmission** *Curr Opin Neurobiol* **29**:25–32 https://doi.org/10.1016/j.conb.2014.04.010 | Google Scholar

[46] Tritsch N.X., Granger A.J., Sabatini B.L. (2016) **Mechanisms and functions of GABA co-release** *Nat Rev Neurosci* **17**:139–45 https://doi.org/10.1038/nrn.2015.21 | Google Scholar

[47] Landry C.F., Ellison J.A., Pribyl T.M., Campagnoni C., Kampf K., Campagnoni A.T. (1996) **Myelin basic protein gene expression in neurons: developmental and regional changes in protein targeting within neuronal nuclei, cell bodies, and processes** *J Neurosci* **16**:2452–62 https://doi.org/10.1523/JNEUROSCI.16-08-02452.1996 | Google Scholar

[48] Lutz D., Loers G., Kleene R., Oezen I., Kataria H., Katagihallimath N., et al. (2014) **Myelin basic protein cleaves cell adhesion molecule L1 and promotes neuritogenesis and cell survival** *J Biol Chem* **289**:13503–18 https://doi.org/10.1074/jbc.M113.530238 | Google Scholar

[49] Cheli V.T., Santiago Gonzalez D.A., Spreuer V., Handley V., Campagnoni A.T., Paez P.M. (2016) **Golli Myelin Basic Proteins Modulate Voltage-Operated Ca(++) Influx and Development in Cortical and Hippocampal Neurons** *Mol Neurobiol* **53**:5749–71 https://doi.org/10.1007/s12035-015-9499-1 | Google Scholar

[50] Melone M., Ciappelloni S., Conti F. (2014) **Plasma membrane transporters GAT-1 and GAT-3 contribute to heterogeneity of GABAergic synapses in neocortex** *Front Neuroanat* **8**:72 https://doi.org/10.3389/fnana.2014.00072 | Google Scholar

[51] Theparambil S.M., Hosford P.S., Ruminot I., Kopach O., Reynolds J.R., Sandoval P.Y., et al. (2020) **Astrocytes regulate brain extracellular pH via a neuronal activity-dependent bicarbonate shuttle** *Nat Commun* **11**:5073 https://doi.org/10.1038/s41467-020-18756-3 | Google Scholar

[52] Kubo Y., Baldwin T.J., Jan Y.N., Jan L.Y. (1993) **Primary structure and functional expression of a mouse inward rectifier potassium channel** *Nature* **362**:127–33 https://doi.org/10.1038/362127a0 | Google Scholar

[53] Burnashev N., Khodorova A., Jonas P., Helm P.J., Wisden W., Monyer H., et al. (1992) **Calcium-Permeable AMPA-Kainate Receptors in Fusiform Cerebellar Glial Cells** *Science* **256**:1566–

70 https://doi.org/10.1126/science.1317970 | Google Scholar

[54] Geiger J.R.P., Melcher T., Koh D.S., Sakmann B., Jonas S.P. H. P, et al. (1995) **Relative Abundance of Subunit mRNAs Determines Gating and Ca*+ Permeability of AMPA Receptors in Principal Neurons and Interneurons in Rat CNS** *Neuron* **15**:193–204 https://doi.org/10.1016/0896-6273(95)90076-4 | Google Scholar

[55] Marques S., Zeisel A., Codeluppi S., van Bruggen D., Mendanha Falcao A., Xiao L., et al. (2016) **Oligodendrocyte heterogeneity in the mouse juvenile and adult central nervous system** *Science* **352**:1326–9 https://doi.org/10.1126/science.aaf6463 | Google Scholar

[56] Kasai M., Omae Y., Kawai Y., Shibata A., Hoshino A., Mizuguchi M., et al. (2022) **GWAS identifies candidate susceptibility loci and microRNA biomarkers for acute encephalopathy with biphasic seizures and late reduced diffusion** *Sci Rep* **12**:1332 https://doi.org/10.1038/s41598-021-04576-y | Google Scholar

[57] DePaula-Silva A.B., Gorbea C., Doty D.J., Libbey J.E., Sanchez J.M.S., Hanak T.J., et al. (2019) **Differential transcriptional profiles identify microglial- and macrophage-specific gene markers expressed during virus-induced neuroinflammation** *J Neuroinflammation* **16**:152 https://doi.org/10.1186/s12974-019-1545-x | Google Scholar

[58] Koutsodendris N., Blumenfeld J., Agrawal A., Traglia M., Grone B., Zilberter M., et al. (2023) **Neuronal APOE4 removal protects against tau-mediated gliosis, neurodegeneration and myelin deficits** *Nat Aging* **3**:275–96 https://doi.org/10.1038/s43587-023-00368-3 | Google Scholar

[59] Schirmer L., Velmeshev D., Holmqvist S., Kaufmann M., Werneburg S., Jung D., et al. (2019) **Neuronal vulnerability and multilineage diversity in multiple sclerosis** *Nature* **573**:75–82 https://doi.org/10.1038/s41586-019-1404-z | Google Scholar

[60] Creus-Muncunill J., Haure-Mirande J.V., Mattei D., Bons J., Ramirez A.V., Hamilton B.W., et al. (2024) **TYROBP/DAP12 knockout in Huntington's disease Q175 mice cell-autonomously decreases microglial expression of disease-associated genes and non-cell-autonomously mitigates astrogliosis and motor deterioration** *J Neuroinflammation* **21**:66 https://doi.org/10.1186/s12974-024-03052-4 | Google Scholar

[61] Tang L.L., Wu Y.B., Fang C.Q., Qu P., Gao Z.L. (2016) **NDRG2 promoted secreted miR-375 in microvesicles shed from M1 microglia, which induced neuron damage** *Biochem Biophys Res Commun* **469**:392–8 https://doi.org/10.1016/j.bbrc.2015.11.098 | Google Scholar

[62] Cheng W., Ndoka E., Hutch C., Roelofs K., MacKinnon A., Khoury B., et al. (2020) **Leptin receptor-expressing nucleus tractus solitarius neurons suppress food intake independently of GLP1 in mice** *JCI Insight* **5** https://doi.org/10.1172/jci.insight.134359 | Google Scholar

[63] Roman C.W., Sloat S.R., Palmiter R.D. (2017) **A tale of two circuits: CCK(NTS) neuron stimulation controls appetite and induces opposing motivational states by projections to distinct brain regions** *Neuroscience* **358**:316–24 https://doi.org/10.1016/j.neuroscience.2017.06.049 | Google Scholar

[64] Chou V., Pearse R.V., Aylward A.J., Ashour N., Taga M., Terzioglu G., et al. (2023) **INPP5D regulates inflammasome activation in human microglia** *Nat Commun* **14**:7552 https://doi.org/10.1038/s41467-023-42819-w | Google Scholar

[65] Kleeman S.O., Thakir T.M., Demestichas B., Mourikis N., Loiero D., Ferrer M., et al. (2023) **Cystatin C is glucocorticoid responsive, directs recruitment of Trem2+ macrophages, and predicts failure of cancer immunotherapy** *Cell Genom* **3**:100347 https://doi.org/10.1016/j.xgen.2023.100347 | Google Scholar

[66] Kang S.H., Fukaya M., Yang J.K., Rothstein J.D., Bergles D.E. (2010) **NG2+ CNS glial progenitors remain committed to the oligodendrocyte lineage in postnatal life and following neurodegeneration** *Neuron* **68**:668–81 https://doi.org/10.1016/j.neuron.2010.09.009 | Google Scholar

[67] Bachor T.P., Hwang E., Yulyaningsih E., Attal K., Mifsud F., Pham V., et al. (2024) **Identification of AgRP cells in the murine hindbrain that drive feeding** *Mol Metab* **80**:101886 https://doi.org/10.1016/j.molmet.2024.101886 | Google Scholar

[68] Holt M.K. (2022) **The ins and outs of the caudal nucleus of the solitary tract: An overview of cellular populations and anatomical connections** *J Neuroendocrinol* **34**:e13132 https://doi.org/10.1111/jne.13132 | Google Scholar

[69] Breen D.M., Kim H., Bennett D., Calle R.A., Collins S., Esquejo R.M., et al. (2020) **GDF-15 Neutralization Alleviates Platinum-Based Chemotherapy-Induced Emesis, Anorexia, and Weight Loss in Mice and Nonhuman Primates** *Cell Metab* **32**:938–50 https://doi.org/10.1016/j.cmet.2020.10.023 | Google Scholar

[70] Tsuzuki K., Ishiuchi S. (2008) **GluR2 expressed by glial fibrillary acidic protein promoter decreases the number of neurons** *Front Biosci* **13**:2784–96 https://doi.org/10.2741/2885 | Google Scholar

[71] Endo F., Kasai A., Soto J.S., Yu X., Qu Z., Hashimoto H., et al. (2022) **Molecular basis of astrocyte diversity and morphology across the CNS in health and disease** *Science* **378** https://doi.org/10.1126/science.adc9020 | Google Scholar

[72] Lujan R. (2010) **Organisation of potassium channels on the neuronal surface** *J Chem Neuroanat* **40**:1–20 https://doi.org/10.1016/j.jchemneu.2010.03.003 | Google Scholar

[73] Cabezas R., Avila M., Gonzalez J., El-Bacha R.S., Baez E., Garcia-Segura L.M., et al. (2014) **Astrocytic modulation of blood brain barrier: perspectives on Parkinson's disease** *Front Cell Neurosci* **8**:211 https://doi.org/10.3389/fncel.2014.00211 | Google Scholar

[74] Cheng W., Gonzalez I., Pan W., Tsang A.H., Adams J., Ndoka E., et al. (2020) **Calcitonin Receptor Neurons in the Mouse Nucleus Tractus Solitarius Control Energy Balance via the Non-aversive Suppression of Feeding** *Cell Metab* **31**:301–12 https://doi.org/10.1016/j.cmet.2019.12.012 | Google Scholar

[75] Patel A.R., Frikke-Schmidt H., Sabatini P.V., Rupp A.C., Sandoval D.A., Myers M.G., et al. (2023) **Neither GLP-1 receptors nor GFRAL neurons are required for aversive or anorectic response to DON (vomitoxin)** *Am J Physiol Regul Integr Comp Physiol* **324**:R635–44 https://doi.org/10.1152/ajpregu.00189.2022 | Google Scholar

[76] Robins S.C., Trudel E., Rotondi O., Liu X., Djogo T., Kryzskaya D., et al. (2013) **Evidence for NG2-glia derived, adult-born functional neurons in the hypothalamus** *PLoS One* **8**:e78236 https:

//doi.org/10.1371/journal.pone.0078236 | Google Scholar

[77] Liberini C.G., Borner T., Boyle C.N., Lutz T.A. (2016) **The satiating hormone amylin enhances neurogenesis in the area postrema of adult rats** *Mol Metab* **5**:834–43 https://doi.org/10.1016/j.molmet.2016.06.015 | Google Scholar

[78] Wada N., Hirako S., Takenoya F., Kageyama H., Okabe M., Shioda S. (2014) **Leptin and its receptors** *J Chem Neuroanat* :61–62 https://doi.org/10.1016/j.jchemneu.2014.09.002 | Google Scholar

[79] Munzberg H., Flier J.S., Bjorbaek C. (2004) **Region-specific leptin resistance within the hypothalamus of diet-induced obese mice** *Endocrinology* **145**:4880–9 https://doi.org/10.1210/en.2004-0726 | Google Scholar

[80] Smith P.M., Brzezinska P., Hubert F., Mimee A., Maurice D.H., Ferguson A.V. (2016) **Leptin influences the excitability of area postrema neurons** *Am J Physiol Regul Integr Comp Physiol* **310**:R440–8 https://doi.org/10.1152/ajpregu.00326.2015 | Google Scholar

[81] Liberini C.G., Boyle C.N., Cifani C., Venniro M., Hope B.T., Lutz T.A. (2016) **Amylin receptor components and the leptin receptor are co-expressed in single rat area postrema neurons** *Eur J Neurosci* **43**:653–61 https://doi.org/10.1111/ejn.13163 | Google Scholar

[82] Huo L., Maeng L., Bjorbaek C., Grill H.J. (2007) **Leptin and the control of food intake: neurons in the nucleus of the solitary tract are activated by both gastric distension and leptin** *Endocrinology* **148**:2189–97 https://doi.org/10.1210/en.2006-1572 | Google Scholar

[83] Wilding J.P.H., Batterham R.L., Calanna S., Davies M., Van Gaal L.F., Lingvay I., et al. (2021) **Once-Weekly Semaglutide in Adults with Overweight or Obesity** *N Engl J Med* **384**:989–1002 https://doi.org/10.1056/NEJMoa2032183 | Google Scholar

[84] Lake B.B., Ai R., Kaeser G.E., Salathia N.S., Yung Y.C., Liu R., et al. (2016) **Neuronal subtypes and diversity revealed by single-nucleus RNA sequencing of the human brain** *Science* **352**:1586–90 https://doi.org/10.1126/science.aaf1204 | Google Scholar

[85] Lake B.B., Codeluppi S., Yung Y.C., Gao D., Chun J., Kharchenko P.V., et al. (2017) **A comparative strategy for single-nucleus and single-cell transcriptomes confirms accuracy in predicted cell-type expression from nuclear RNA** *Sci Rep* **7**:6031 https://doi.org/10.1038/s41598-017-04426-w | Google Scholar

[86] Thrupp N., Sala Frigerio C., Wolfs L., Skene N.G., Fattorelli N., Poovathingal S., et al. (2020) **Single-Nucleus RNA-Seq Is Not Suitable for Detection of Microglial Activation Genes in Humans** *Cell Rep* **32**:108189 https://doi.org/10.1016/j.celrep.2020.108189 | Google Scholar

[87] Thorvaldsdottir H., Robinson J.T., Mesirov J.P. (2013) **Integrative Genomics Viewer (IGV): high-performance genomics data visualization and exploration** *Brief Bioinform* **14**:178–92 https://doi.org/10.1093/bib/bbs017 | Google Scholar

[88] Park M.J., Chung K. (1999) **Endogenous lectin (RL-29) expression of the autonomic preganglionic neurons in the rat spinal cord** *The Anatomical Record* **254**:54–60 https://doi.org/10.1002/(SICI)1097-0185(19990101)254:1<53::AID-AR7>3.0.CO;2-H | Google Scholar

[89] Miranda C.O., Hegedus K., Wildner H., Zeilhofer H.U., Antal M. (2022) **Morphological and neurochemical characterization of glycinergic neurons in laminae I-IV of the mouse**

spinal dorsal horn *J Comp Neurol* **530**:607–26 https://doi.org/10.1002/cne.25232 | Google Scholar

[90] Guo F., Wang Y. (2023) **TCF7l2, a nuclear marker that labels premyelinating oligodendrocytes and promotes oligodendroglial lineage progression** *Glia* **71**:143–54 https://doi.org/10.1002/glia.24249 | Google Scholar

[91] Weihe E., Depboylu C., Schutz B., Schafer M.K., Eiden L.E. (2006) **Three types of tyrosine hydroxylase-positive CNS neurons distinguished by dopa decarboxylase and VMAT2 co-expression** *Cell Mol Neurobiol* **26**:659–78 https://doi.org/10.1007/s10571-006-9053-9 | Google Scholar

[92] Sun J., Song Y., Chen Z., Qiu J., Zhu S., Wu L., et al. (2022) **Heterogeneity and Molecular Markers for CNS Glial Cells Revealed by Single-Cell Transcriptomics** *Cell Mol Neurobiol* **42**:2629–42 https://doi.org/10.1007/s10571-021-01159-3 | Google Scholar

[93] Dallaporta M., Bonnet M.S., Horner K., Trouslard J., Jean A., Troadec J.D. (2010) **Glial cells of the nucleus tractus solitarius as partners of the dorsal hindbrain regulation of energy balance: a proposal for a working hypothesis** *Brain Res* **1350**:35–42 https://doi.org/10.1016/j.brainres.2010.04.025 | Google Scholar

[94] Kalucka J., de Rooij L., Goveia J., Rohlenova K., Dumas S.J., Meta E., et al. (2020) **Single-Cell Transcriptome Atlas of Murine Endothelial Cells** *Cell* **180**:764–79 https://doi.org/10.1016/j.cell.2020.01.015 | Google Scholar

[95] Dani N., Herbst R.H., McCabe C., Green G.S., Kaiser K., Head J.P., et al. (2021) **A cellular and spatial map of the choroid plexus across brain ventricles and ages** *Cell* **184**:3056–74 https://doi.org/10.1016/j.cell.2021.04.003 | Google Scholar

[96] Ray B., Bailey J.A., Sarkar S., Lahiri D.K. (2009) **Molecular and immunocytochemical characterization of primary neuronal cultures from adult rat brain: Differential expression of neuronal and glial protein markers** *J Neurosci Methods* **184**:294–302 https://doi.org/10.1016/j.jneumeth.2009.08.018 | Google Scholar

[97] Lauri S.E., Ryazantseva M., Orav E., Vesikansa A., Taira T. (2021) **Kainate receptors in the developing neuronal networks** *Neuropharmacology* **195**:108585 https://doi.org/10.1016/j.neuropharm.2021.108585 | Google Scholar

[98] Song Y.H., Yoon J., Lee S.H. (2021) **The role of neuropeptide somatostatin in the brain and its application in treating neurological disorders** *Exp Mol Med* **53**:328–38 https://doi.org/10.1038/s12276-021-00580-4 | Google Scholar

[99] Song H.W., Foreman K.L., Gastfriend B.D., Kuo J.S., Palecek S.P., Shusta E.V. (2020) **Transcriptomic comparison of human and mouse brain microvessels** *Sci Rep* **10**:12358 https://doi.org/10.1038/s41598-020-69096-7 | Google Scholar

[100] Xu L., Yao Y. (2021) **Central Nervous System Fibroblast-Like Cells in Stroke and Other Neurological Disorders** *Stroke* **52**:2456–64 https://doi.org/10.1161/STROKEAHA.120.033431 | Google Scholar

[101] Ernsberger U., Rohrer H. (2018) **Sympathetic tales: subdivisons of the autonomic nervous system and the impact of developmental studies** *Neural Dev* **13**:20 https://doi.org/10.1186/s13064-018-0117-6 | Google Scholar

[102] Gajera C.R., Emich H., Lioubinski O., Christ A., Beckervordersandforth-Bonk R., Yoshikawa K., et al. (2010) **LRP2 in ependymal cells regulates BMP signaling in the adult neurogenic niche** *J Cell Sci* **123**:1922–30 https://doi.org/10.1242/jcs.065912 | Google Scholar

[103] MacDonald A., Lu B., Caron M., Caporicci-Dinucci N., Hatrock D., Petrecca K., et al. (2021) **Single Cell Transcriptomics of Ependymal Cells Across Age, Region and Species Reveals Cilia-Related and Metal Ion Regulatory Roles as Major Conserved Ependymal Cell Functions** *Front Cell Neurosci* **15**:703951 https://doi.org/10.3389/fncel.2021.703951 | Google Scholar

[104] Jeong H.W., Dieguez-Hurtado R., Arf H., Song J., Park H., Kruse K., et al. (2022) **Single-cell transcriptomics reveals functionally specialized vascular endothelium in brain** *eLife* **11** https://doi.org/10.7554/eLife.57520 | Google Scholar

[105] Huizer K., Mustafa D.A.M., Spelt J.C., Kros J.M., Sacchetti A. (2017) **Improving the characterization of endothelial progenitor cell subsets by an optimized FACS protocol** *PLoS One* **12**:e0184895 https://doi.org/10.1371/journal.pone.0184895 | Google Scholar

[106] Gerrits E., Heng Y., Boddeke E., Eggen B.J.L. (2020) **Transcriptional profiling of microglia; current state of the art and future perspectives** *Glia* **68**:740–55 https://doi.org/10.1002/glia.23767 | Google Scholar

[107] Shah P.T., Stratton J.A., Stykel M.G., Abbasi S., Sharma S., Mayr K.A., et al. (2018) **Single-Cell Transcriptomics and Fate Mapping of Ependymal Cells Reveals an Absence of Neural Stem Cell Function** *Cell* **173**:1045–57 https://doi.org/10.1016/j.cell.2018.03.063 | Google Scholar

[108] Causing C.G., Gloster A., Aloyz R., Bamji S.X., Chang E., Fawcett J., et al. (1997) **Synaptic Innervation Density Is Regulated by Neuron-Derived BDNF** *Neuron* **18**:257–67 https://www.cell.com/fulltext/S0896-6273(00)80266-4#secd33944447e474 | Google Scholar

[109] Zhang Y., Sloan S.A., Clarke L.E., Caneda C., Plaza C.A., Blumenthal P.D., et al. (2016) **Purification and Characterization of Progenitor and Mature Human Astrocytes Reveals Transcriptional and Functional Differences with Mouse** *Neuron* **89**:37–53 https://doi.org/10.1016/j.neuron.2015.11.013 | Google Scholar

[110] Langlet F. (2019) **Tanycyte Gene Expression Dynamics in the Regulation of Energy Homeostasis** *Front Endocrinol* **10**:286 https://doi.org/10.3389/fendo.2019.00286 | Google Scholar

[111] Uriarte Huarte O., Kyriakis D., Heurtaux T., Pires-Afonso Y., Grzyb K., Halder R., et al. (2021) **Single-Cell Transcriptomics and In Situ Morphological Analyses Reveal Microglia Heterogeneity Across the Nigrostriatal Pathway** *Front Immunol* **12**:639613 https://doi.org/10.3389/fimmu.2021.639613 | Google Scholar

[112] Pepper R.E., Pitman K.A., Cullen C.L., Young K.M. (2018) **How Do Cells of the Oligodendrocyte Lineage Affect Neuronal Circuits to Influence Motor Function, Memory and Mood?** *Front Cell Neurosci* **12**:399 https://doi.org/10.3389/fncel.2018.00399 | Google Scholar

[113] Maegawa A., Murata K., Kuroda K., Fujieda S., Fukazawa Y. (2022) **Cellular Profiles of Prodynorphin and Preproenkephalin mRNA-Expressing Neurons in the Anterior Olfactory

Tubercle of Mice *Front Neural Circuits* **16**:908964 https://doi.org/10.3389/fncir.2022.908964 | Google Scholar

[114] Hentges S.T., Otero-Corchon V., Pennock R.L., King C.M., Low M.J. (2009) **Proopiomelanocortin expression in both GABA and glutamate neurons** *J Neurosci* **29**:13684–90 https://doi.org/10.1523/JNEUROSCI.3770-09.2009 | Google Scholar

[115] Sun W., Cornwell A., Li J., Peng S., Osorio M.J., Aalling N., et al. (2017) **SOX9 Is an Astrocyte-Specific Nuclear Marker in the Adult Brain Outside the Neurogenic Regions** *J Neurosci* **37**:4493–507 https://doi.org/10.1523/JNEUROSCI.3199-16.2017 | Google Scholar

[116] Lee A.T., Vogt D., Rubenstein J.L., Sohal V.S. (2014) **A class of GABAergic neurons in the prefrontal cortex sends long-range projections to the nucleus accumbens and elicits acute avoidance behavior** *J Neurosci* **34**:11519–25 https://doi.org/10.1523/JNEUROSCI.1157-14.2014 | Google Scholar

[117] Vanlandewijck M., He L., Mae M.A., Andrae J., Ando K., Del Gaudio F., et al. (2018) **A molecular atlas of cell types and zonation in the brain vasculature** *Nature* **554**:475–80 https://doi.org/10.1038/nature25739 | Google Scholar

[118] Carlen M., Meletis K., Goritz C., Darsalia V., Evergren E., Tanigaki K., et al. (2009) **Forebrain ependymal cells are Notch-dependent and generate neuroblasts and astrocytes after stroke** *Nat Neurosci* **12**:259–67 https://doi.org/10.1038/nn.2268 | Google Scholar

[119] Garcia-Revilla J., Boza-Serrano A., Espinosa-Oliva A.M., Soto M.S., Deierborg T., Ruiz R., et al. (2022) **Galectin-3, a rising star in modulating microglia activation under conditions of neurodegeneration** *Cell Death Dis* **13**:628 https://doi.org/10.1038/s41419-022-05058-3 | Google Scholar

[120] Yang A.C., Vest R.T., Kern F., Lee D.P., Agam M., Maat C.A., et al. (2022) **A human brain vascular atlas reveals diverse mediators of Alzheimer's risk** *Nature* **603**:885–92 https://doi.org/10.1038/s41586-021-04369-3 | Google Scholar

[121] Chasseigneaux S., Moraca Y., Cochois-Guegan V., Boulay A.C., Gilbert A., Le Crom S., et al. (2018) **Isolation and differential transcriptome of vascular smooth muscle cells and mid-capillary pericytes from the rat brain** *Sci Rep* **8**:12272 https://doi.org/10.1038/s41598-018-30739-5 | Google Scholar

[122] Gao Z., Ure K., Ables J.L., Lagace D.C., Nave K.A., Goebbels S., et al. (2009) **Neurod1 is essential for the survival and maturation of adult-born neurons** *Nat Neurosci* **12**:1090–2 https://doi.org/10.1038/nn.2385 | Google Scholar

[123] Kodama T., Guerrero S., Shin M., Moghadam S., Faulstich M., du Lac S. (2012) **Neuronal classification and marker gene identification via single-cell expression profiling of brainstem vestibular neurons subserving cerebellar learning** *J Neurosci* **32**:7819–31 https://doi.org/10.1523/JNEUROSCI.0543-12.2012 | Google Scholar

[124] Manjarrez-Marmolejo J., Franco-Pérez J. (2016) **Gap Junction Blockers-An Overview of their Effects on Induced Seizures in Animal Models** *Current Neuropharmacology* **14**:759–71 https://doi.org/10.2174/1570159X14666160603115 | Google Scholar

[125] Paretkar T., Dimitrov E. (2019) **Activation of enkephalinergic (Enk) interneurons in the central amygdala (CeA) buffers the behavioral effects of persistent pain** *Neurobiol Dis* **124**:364–72 https://doi.org/10.1016/j.nbd.2018.12.005 | Google Scholar

[126] Zhang Z., Ma Z., Zou W., Guo H., Liu M., Ma Y., et al. (2019) **The Appropriate Marker for Astrocytes: Comparing the Distribution and Expression of Three Astrocytic Markers in Different Mouse Cerebral Regions** *Biomed Res Int* **9605265** https://doi.org/10.1155/2019/9605265 | Google Scholar

[127] Jurga A.M., Paleczna M., Kadluczka J., Kuter K.Z. (2021) **Beyond the GFAP-Astrocyte Protein Markers in the Brain** *Biomolecules* **11** https://doi.org/10.3390/biom11091361 | Google Scholar

[128] Dyachuk V., Furlan A., Shahidi M.K., Giovenco M., Kaukua N., Konstantinidou C., et al. (2014) **Parasympathetic neurons originate from nerve-associated peripheral glial progenitors** *Science* **345**:82–7 https://doi.org/10.1126/science.1253281 | Google Scholar

[129] Liu D.S., Xu T.L. (2019) **Cell-Type Identification in the Autonomic Nervous System** *Neurosci Bull* **35**:145–55 https://doi.org/10.1007/s12264-018-0284-9 | Google Scholar

[130] Kang B.J., Chang D.A., Mackay D.D., West G.H., Moreira T.S., Takakura A.C., et al. (2007) **Central nervous system distribution of the transcription factor Phox2b in the adult rat** *J Comp Neurol* **503**:627–41 https://doi.org/10.1002/cne.21409 | Google Scholar

[131] Adler M., Moriel N., Goeva A., Avraham-Davidi I., Mages S., Adams T.S., et al. (2023) **Emergence of division of labor in tissues through cell interactions and spatial cues** *Cell Rep* **42**:112412 https://doi.org/10.1016/j.celrep.2023.112412 | Google Scholar

[132] Steimel A., Suh J., Hussainkhel A., Deheshi S., Grants J.M., Zapf R., et al. (2013) **The C. elegans CDK8 Mediator module regulates axon guidance decisions in the ventral nerve cord and during dorsal axon navigation** *Dev Biol* **377**:385–98 https://doi.org/10.1016/j.ydbio.2013.02.009 | Google Scholar

[133] Daiber T., VanderZwan-Butler C.J., Bashaw G.J., Evans T.A. (2021) **Conserved and divergent aspects of Robo receptor signaling and regulation between Drosophila Robo1 and C. elegans SAX-3** *Genetics* **217** https://doi.org/10.1093/genetics/iyab018 | Google Scholar

[134] Schmetsdorf S., Gärtner U., Arendt T. (2007) **Constitutive Expression of Functionally Active Cyclin-Dependent Kinases and Their Binding Partners Suggests Noncanonical Functions of Cell Cycle Regulators in Differentiated Neurons** *Cerebral Cortex* **17**:1821–9 https://doi.org/10.1093/cercor/bhl091 | Google Scholar

[135] Kummer K.K., Zeidler M., Kalpachidou T., Kress M. (2021) **Role of IL-6 in the regulation of neuronal development, survival and function** *Cytokine* **144**:155582 https://doi.org/10.1016/j.cyto.2021.155582 | Google Scholar

[136] März P., Cheng J.G., Gadient R.A., Patterson P.H., Stoyan T., Otten U., et al. (1998) **Sympathetic neurons can produce and respond to interleukin 6** *Proc Natl Acad Sci* **95**:3251–6 https://doi.org/10.1073/pnas.95.6.3251 | Google Scholar

[137] McGrail K.M., Phillips J.M., Sweadner K.J. (1991) **Immunofluorescent localization of three Na,K-ATPase isozymes in the rat central nervous system: both neurons and glia can express more than one Na,K-ATPase** *J Neurosci* **11**:381–91 https://doi.org/10.1523/JNEUROSCI.11-02-00381.1991 | Google Scholar

[138] Poulsen H., Khandelia H., Morth J.P., Bublitz M., Mouritsen O.G., Egebjerg J., et al. (2010) **Neurological disease mutations compromise a C-terminal ion pathway in the Na(+)/K(+)-ATPase** *Nature* **467**:99–102 https://doi.org/10.1038/nature09309 | Google Scholar

[139] Abousaab A., Warsi J., Elvira B., Lang F. (2016) **Caveolin-1 Sensitivity of Excitatory Amino Acid Transporters EAAT1, EAAT2, EAAT3, and EAAT4** *J Membr Biol* **249**:239–49 https://doi.org/10.1007/s00232-015-9863-0 | Google Scholar

[140] Bongarzone E.R., Campagnoni C.W., Kampf K., Jacobs E.C., Handley V.W., Schonmann V., et al. (1999) **Identification of a new exon in the myelin proteolipid protein gene encoding novel protein isoforms that are restricted to the somata of oligodendrocytes and neurons** *J Neurosci* **19**:8349–57 https://doi.org/10.1523/JNEUROSCI.19-19-08349 | Google Scholar

[141] Nunnelly L.F., Campbell M., Lee D.I., Dummer P., Gu G., Menon V., et al. (2022) **St18 specifies globus pallidus projection neuron identity in MGE lineage** *Nat Commun* **13**:7735 https://doi.org/10.1038/s41467-022-35518-5 | Google Scholar

[142] Plachez C., Lindwall C., Sunn N., Piper M., Moldrich R.X., Campbell C.E., et al. (2008) **Nuclear factor I gene expression in the developing forebrain** *J Comp Neurol* **508**:385–401 https://doi.org/10.1002/cne.21645 | Google Scholar

[143] Fraser J., Essebier A., Brown A.S., Davila R.A., Harkins D., Zalucki O., et al. (2020) **Common Regulatory Targets of NFIA, NFIX and NFIB during Postnatal Cerebellar Development** *Cerebellum* **19**:89–101 https://doi.org/10.1007/s12311-019-01089-3 | Google Scholar

[144] Dubois L., Vincent A. (2001) **The COE – Collier/Olf1/EBF – transcription factors: structural conservation and diversity of developmental functions** *Mechanisms of Development* **108**:3–12 https://doi.org/10.1016/s0925-4773(01)00486-5 | Google Scholar

[145] Liu Y., Lear T., Iannone O., Shiva S., Corey C., Rajbhandari S., et al. (2015) **The Proapoptotic F-box Protein Fbxl7 Regulates Mitochondrial Function by Mediating the Ubiquitylation and Proteasomal Degradation of Survivin** *J Biol Chem* **290**:11843–52 https://doi.org/10.1074/jbc.M114.629931 | Google Scholar

[146] Kim H., Berens N.C., Ochandarena N.E., Philpot B.D. (2020) **Region and Cell Type Distribution of TCF4 in the Postnatal Mouse Brain** *Front Neuroanat* **14**:42 https://doi.org/10.3389/fnana.2020.00042 | Google Scholar

[147] Sirp A., Shubina A., Tuvikene J., Tamberg L., Kiir C.S., Kranich L., et al. (2022) **Expression of alternative transcription factor 4 mRNAs and protein isoforms in the developing and adult rodent and human tissues** *Front Mol Neurosci* **15**:1033224 https://doi.org/10.3389/fnmol.2022.1033224 | Google Scholar

[148] Hodge R.D., Bakken T.E., Miller J.A., Smith K.A., Barkan E.R., Graybuck L.T., et al. (2019) **Conserved cell types with divergent features in human versus mouse cortex** *Nature* **573**:61–8 https://doi.org/10.1038/s41586-019-1506-7 | Google Scholar

[149] Kuroda T., Yasuda S., Tachi S., Matsuyama S., Kusakawa S., Tano K., et al. (2019) **SALL3 expression balance underlies lineage biases in human induced pluripotent stem cell differentiation** *Nat Commun* **10**:2175 https://doi.org/10.1038/s41467-019-09511-4 | Google Scholar

[150] Morello F., Borshagovski D., Survila M., Tikker L., Sadik-Ogli S., Kirjavainen A., et al. (2020) **Molecular Fingerprint and Developmental Regulation of the Tegmental GABAergic and**

Glutamatergic Neurons Derived from the Anterior Hindbrain *Cell Rep* **33**:108268 https://doi.org/10.1016/j.celrep.2020.108268 | Google Scholar

[151] Wolf L., Gao C.S., Gueta K., Xie Q., Chevallier T., Podduturi N.R., et al. (2013) **Identification and characterization of FGF2-dependent mRNA: microRNA networks during lens fiber cell differentiation** *G3 (Bethesda)* **3**:2239–55 https://doi.org/10.1534/g3.113.008698 | Google Scholar

[152] Liu H., Aramaki M., Fu Y., Forrest D. (2017) **Retinoid-Related Orphan Receptor beta and Transcriptional Control of Neuronal Differentiation** *Curr Top Dev Biol* **125**:227–55 https://doi.org/10.1016/bs.ctdb.2016.11.009 | Google Scholar

[153] Del Barrio M.G., Bourane S., Grossmann K., Schule R., Britsch S., O'Leary D.D., et al. (2013) **A transcription factor code defines nine sensory interneuron subtypes in the mechanosensory area of the spinal cord** *PLoS One* **8**:e77928 https://doi.org/10.1371/journal.pone.0077928 | Google Scholar

[154] Li Y., Osuma A., Correa E., Okebalama M.A., Dao P., Gaylord O., et al. (2020) **Establishment and maintenance of motor neuron identity via temporal modularity in terminal selector function** *eLife* **9** https://doi.org/10.7554/eLife.59464 | Google Scholar

[155] Moruzzo D., Nobbio L., Sterlini B., Consalez G.G., Benfenati F., Schenone A., et al. (2017) **The Transcription Factors EBF1 and EBF2 Are Positive Regulators of Myelination in Schwann Cells** *Mol Neurobiol* **54**:8117–27 https://doi.org/10.1007/s12035-016-0296-2 | Google Scholar

[156] Jimenez M.A., Akerblad P., Sigvardsson M., Rosen E.D. (2007) **Critical role for Ebf1 and Ebf2 in the adipogenic transcriptional cascade** *Mol Cell Biol* **27**:743–57 https://doi.org/10.1128/MCB.01557-06 | Google Scholar

[157] Yu H., Rubinstein M., Low M.J. (2022) **Developmental single-cell transcriptomics of hypothalamic POMC neurons reveal the genetic trajectories of multiple neuropeptidergic phenotypes** *eLife* **11** https://doi.org/10.7554/eLife.72883 | Google Scholar

[158] Singh I., Wang L., Xia B., Liu J., Tahiri A., El Ouaamari A., et al. (2022) **Activation of arcuate nucleus glucagon-like peptide-1 receptor-expressing neurons suppresses food intake** *Cell Biosci* **12**:178 https://doi.org/10.1186/s13578-022-00914-3 | Google Scholar

[159] Xiao L., Merullo D.P., Koch T.M.I., Cao M., Co M., Kulkarni A., et al. (2021) **Expression of FoxP2 in the basal ganglia regulates vocal motor sequences in the adult songbird** *Nat Commun* **12**:2617 https://doi.org/10.1038/s41467-021-22918-2 | Google Scholar

[160] Du T.T., Dewey J.B., Wagner E.L., Cui R., Heo J., Park J.J., et al. (2019) **LMO7 deficiency reveals the significance of the cuticular plate for hearing function** *Nat Commun* **10**:1117 https://doi.org/10.1038/s41467-019-09074-4 | Google Scholar

[161] Wozniak M.A., Baker B.M., Chen C.S., Wilson K.L. (2013) **The emerin-binding transcription factor Lmo7 is regulated by association with p130Cas at focal adhesions** *PeerJ* **1**:e134 https://doi.org/10.7717/peerj.134 | Google Scholar

[162] Kawaue T., Shitamukai A., Nagasaka A., Tsunekawa Y., Shinoda T., Saito K., et al. (2019) **Lzts1 controls both neuronal delamination and outer radial glial-like cell generation during**

mammalian cerebral development *Nat Commun* **10**:2780 https://doi.org/10.1038/s41467-019-10730-y | Google Scholar

[163] Randlett O., Poggi L., Zolessi F.R., Harris W.A. (2011) **The oriented emergence of axons from retinal ganglion cells is directed by laminin contact in vivo** *Neuron* **70**:266–80 https://doi.org/10.1016/j.neuron.2011.03.013 | Google Scholar

[164] Ichikawa-Tomikawa N., Ogawa J., Douet V., Xu Z., Kamikubo Y., Sakurai T., et al. (2012) **Laminin alpha1 is essential for mouse cerebellar development** *Matrix Biol* **31**:17–28 https://doi.org/10.1016/j.matbio.2011.09.002 | Google Scholar

[165] Liau E.S., Jin S., Chen Y.C., Liu W.S., Calon M., Nedelec S., et al. (2023) **Single-cell transcriptomic analysis reveals diversity within mammalian spinal motor neurons** *Nat Commun* **14**:46 https://doi.org/10.1038/s41467-022-35574-x | Google Scholar

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Reviewer #1 (Public review):

Summary:

This paper is using state-of-the-art techniques to define the cellular composition and its complexity in two rodent species (mice and rats). The study is built on available datasets but extends those in a way that future research will be facilitated. The study will be of high impact for the study of metabolic control.

Strengths:

After revision, the paper is much improved. I have no further comments.

https://doi.org/10.7554/eLife.106217.2.sa2

Reviewer #2 (Public review):

In this manuscript, Hes et al. present a comprehensive multi-species atlas of the dorsal vagal complex (DVC) using single-nucleus RNA sequencing, identifying over 180,000 cells and 123 cell types across five levels of granularity in mice and rats. Intriguingly, the analysis uncovered previously uncharacterized cell populations, including Kcnj3-expressing astrocytes, neurons co-expressing Th and Cck, and a population of leptin receptor-expressing neurons in the rat area postrema, which also express the progenitor marker Pdgfra. These findings suggest species-specific differences in appetite regulation. This study provides a valuable resource for investigating the intricate cellular landscape of the DVC and its role in metabolic control, with potential implications for refining obesity treatments targeting this hindbrain region.

In line with previous work published by the PI, the topic is of clear scientific relevance, and the data presented in this manuscript are both novel and compelling. Additionally, the manuscript is well-structured, and the conclusions are robust and supported by the data. Overall, this study significantly enhances our understanding of the DVC and sheds light on key differences between rats and mice.

I have reviewed the revised manuscript and am pleased to confirm that the authors have addressed my previous comments and concerns.

https://doi.org/10.7554/eLife.106217.2.sa1

Author response:

The following is the authors' response to the original reviews

We thank the expert reviewers for their careful consideration of our manuscript and the feedback to help us strengthen our work. Please find a response to each reviewer's comments below. We have included the original text from the reviewer in unbolded text and our response, immediately below, in bold text for clarity.

> ***Reviewer #1:***
>
> *(1) Appetite is controlled, not regulated; please reword throughout.*

The reviewer raises a valid point that we have misused the word "regulate" in certain instances and "control" would be more accurate term. We have made adjustments throughout the manuscript.

(2) One minor point that would further strengthen the data is a more distinct analysis of receptors that are characteristic of the different populations of neuronal and non-neuronal cells; this part could be improved.

We thank the reviewer for this suggestion as we had not directly compared metabolicallyrelevant peptides/receptors between the mouse and rat DVC. We have included a list of selected receptors and neuropeptides expression (see Figure S13) for neuronal cells in mouse and rat. We have included this figure as a new supplement. There are some interesting insights from this data, including the relatively broad expression of Lepr in the rat compared with the mouse and the absence of proglucagon expressing neurons within the rat DVC.

> ***Reviewer #2:***
>
> *(1) In some of the graphs, the label AP/NTS is used, but DVC would be more appropriate.*

We have reviewed the figures and legends to ensure appropriate use of DVC. We thank the reviewer for bringing this oversight to our attention.

> *(2) Line 124, p7 - Sprague Dawley RATS*

We have changed the text to "Sprague Dawley rats"

> *(3) Line 132, p7 - The phrase "were provided with given access to food" needs grammatical correction.*

We agree the text was poorly written. The sentence has been corrected to: "Wild-type Sprague

Dawley rats (Charles River) were provided with ad libitum access to food (Purina Lab Diet

1. and water in temperature-controlled (22°C) rooms on a 12-hour light-dark cycle with daily health checks." We have also reviewed the entire manuscript and made additional amendments where necessary.

> *(4) Page 15 - Mention that GFAP is a marker for astrocytes. Additionally, correct the typo "gfrap".*

We have corrected the misspelling of “Gfap” within the text. We appreciate the reviewer’s comment that there is value in communicating to the nonexpert reader that GFAP is a marker for astrocytes, however, as our data and that from other snRNA-Seq studies show that Gfap mRNA only labels a subset of astrocytes, our preference is to refrain from stating this. Our data suggests the sole use of Gfap as an astrocyte marker will not reflect the true astrocyte population.

> *(5) Line 432, p15 - What was the rationale for selecting clusters 23, 26, and 27?*

We chose to perform subclustering on these clusters because they displayed multiple cell identities when surveyed for the 473 marker genes as described in Methods 2.6. In order to separate these, the granularity was increased in them by sub-clustering.

> *(6) Line 533, p18 - only 5 out of 34 neurons express GFRAL, which makes the language used a little bit misleading. As per the comment above, I would specify that only a subset (X%) of neurons express GFRAL, and apply the same approach for other markers.*

We thank the reviewer for raising this point. We agree the text, as written, was an oversimplification. We adjusted the text as recommended: that a subset (~15%) express detectable Gfral mRNA but is likely an underrepresentation due to the challenges in detecting lowly expressed transcripts such as *Gfral*.

> *(7) Line 547, p18 - This statement appears to refer to rat data specifically, rather than rodent data in general.*

The text has been corrected.

> *(8) Section 3.6 - The discussion on meal-related transcriptional programs in the murine DVC does not mention Figure S10A and B.*

We thank the reviewer for the observation. It is true that we do not discuss this figure. Fig10S is the integration of samples in treeArches, a necessary step to build the hierarchy in python so the learning algorithm uses only genes that are related to identity and not treatment, we obtained the same overlap of samples when we used R to assign identities. This figure demonstrates our integration was successful because it is only considering genes that are not-treatment related to establish identities, those which are expressed by cells regardless of their response to any treatment. For the meal-related analysis, we were interested in the genes that are changed by treatment, and this is why the analysis differed. We have included a sentence in the methods to clarify this point that states: " This sample integration was done to ensure that inter-sample variations were removed for the cell identity steps."

> *(9) Page 5, citation 10 - the author cited a clinical trial for glucagon and GLP-1 receptor dual agonist survodutide for "DVC neurons' role in appetite and energy balance stems from their role as therapeutic targets for obesity". A more appropriate citation (such as a review) would be preferable.*

We appreciate the suggestion by the reviewer. We have updated our references to reflect a recent manuscript from the Alhadeff group which demonstrates the DVC acts as the target of GLP1-based therapies. We have also included a review as suggested 10.1038/s42255-02200606-9.

> *(10) Line 52, p5 - a citation of obesity is needed, as the current ref only pertains to cancer cachexia.*

We have included a reference for obesity.

> *(11) In the discussion, it would be valuable to elaborate on the potential significance of DVCspecific glial cells (perhaps at the end of the second paragraph?).*

We thank the reviewer for this suggestion. Our discovery of a DVC-specific astrocyte transcriptional profile was underrepresented within the discussion. We have attempted to expand this discussion on the suspected roles for these DVC-specific astrocytes. Much of this discussion is based on the distinct localization pattern of Gfap mRNA in the DVC (see Image on Allen Brain ISH) which shows dense signal at the boundary of the AP and NTS. As astrocytes have well established roles in maintaining BBB integrity, it is our speculation that this is a major role of these cells. However, functional studies will be critical to assess the roles of these astrocytes in DVC biology.

> *(12) Line 683, p22 - Consider adding PMID: 38987598 which describes the dissociable GLP-1R circuits.*

We appreciate this recommendation – we have included this reference.

> *(13) The authors suggest that a possible explanation for the discrepancy between snRNA-Seq and in situ hybridization data is that Agrp and Hcrt mRNA reads in snRNA-Seq overwhelmingly mapped to non-coding regions. To what extent could this limitation affect other genes included in the current analyzed 10x datasets?*

As shown by Pool and cols. (https://doi.org/10.1038/s41592-023-02003-w) including intronic reads improves sensitivity and more accurately reflects endogenous gene expression. Therefore, including intronic reads is considered more of a strength than a limitation and is now default in platforms such as CellRanger. While including intronic reads for mapping snRNA-Seq data, we would advise corroboration of snRNA-Seq findings with published literature or detection of coding mRNA or protein. In our case, the detection of hypothalamic neuropeptide via snRNA-Seq data could not be verified by performing in situ hybridizations using probes that detect exons. Therefore, *Hcrt* and *Agrp* having only intronic reads suggest a regulatory (reviewed in https://doi.org/10.3389/fgene.2018.00672) rather than a coding role in the DVC.

> *(14) Given the manuscript's focus on feeding and metabolism, I believe a more detailed description and comparison of the transcription profile of known receptors, neurotransmitters, and neuropeptides involved in food intake and energy homeostasis between mice and rats would add value. Adding a curated list of key genes related to feeding regulation would be particularly informative.*

A similar request was made by reviewer #1. Please see the full response above. Briefly, we have performed additional analysis of the mouse and rat DVC data and included this data as an additional supplemental figure (Figure S13).

> *(15) Line 479-482, p17 - It would be helpful if the authors could quantify (e.g., number and/or percentage) the extent of TH and CCK co-expression.*

We have amended the text of the manuscript to include quantification of Cck and Th colocalization. According to our snRNA-seq data, out of the 764 *Th*-expressing neurons, 80 coexpress *Cck* in the mouse (~10%). The *Cck*-expressing cells are more numerous, 3,821 in total.

> *(16) The number of animals used differs significantly between species, which the authors acknowledge as a limitation in the discussion. Since the authors took advantage of previously published mouse data sets (Ludwig and Dowsett data sets), I wonder if the authors could compare/integrate any rat data set currently available in rats as well to partially address the sample size disparity.*

We agree with the review that our rat database is considerably smaller than our mouse database, making comparisons between rat and mouse DVC challenging. We attempted to increase the size of our rat DVC atlas by incorporating publicly available rat DVC snRNA-Seq data (Reiner et al 2022). However, we found several issues with the quality of this data including low UMIs/cell and gene #/cell. For these reasons, we decided against merging these two datasets. So while relatively small, our rat DVC atlas uses high quality data and serves as a valuable starting point. By introducing TreeArches as a method to relatively easily incorporate new snRNA-Seq data into our own, it is our hope that future studies will do so and thus expand the rat DVC atlas we have built.

> *(17) In the Materials and Methods section, LiCl is mentioned as one of the treatment conditions; however, very little corresponding data are presented or discussed. Please include these results and elaborate on the rationale for selecting LiCl over other anorectic compounds.*

The reviewer is correct, some of the tissues used in this study were from animals treated with LiCl prior to euthanasia. Our intent was to contrast the transcriptional effects induced by LiCl (an anorectic agent with aversive properties) with refeeding (a naturally rewarding and satiating stimuli). However, upon analyzing the data, we found very few transcriptional changes induced by LiCl. It is unclear to us whether this was a technical failure in the experiment and so did not elaborate on the results.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) The use of both sexes is indicated in the discussion, but methods and results do not address sex distribution in the investigated groups. Also, the groups could be more clearly described, e.g., the size of the 2 hour refeeding mouse group varies from n=10 to n=5.*

We have clarified the text, in line with the reviewer's suggestion. There were two cohorts of fasted/ refed mice (n=5 each), so in the manuscript methods it is stated as n=10 because of this. The fasted-only group, which was not refed before euthanasia is a separate group, n=5.

> *(2) Page 20, the last sentence needs to be reworded.*

We thank the reviewer for this recommendation. The text has been amended to improve clarity of the sentence.

> *(3) Page 22, lines 691-692 - this sentence needs to be reworded.*

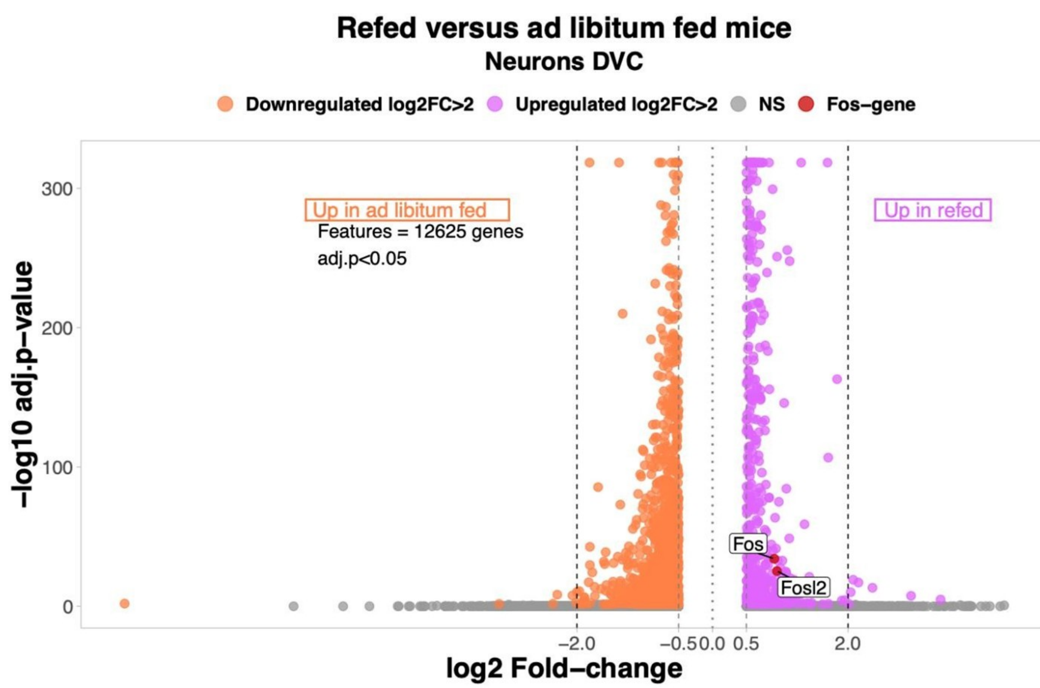
We thank the reviewer for this comment. The offending sentences have been amended.

> *(4) While the authors find transcriptional changes in all neuronal and non-neuronal cell types, which is interesting, the verification of known transcriptional changes (e.g., cFos) is unaddressed. cFos is a common gene upregulated with refeeding that was surprisingly*

> *not investigated, even though this should be a strong maker of proper meal-induced neuronal activation in the DMV. This is a missed opportunity either to verify the data set or to highlight important limitations if that had been attempted without success.*

This is a highly salient point made by the reviewer. Including Fos expression serves as an internal validation of our refeeding condition and the absence of Fos mRNA levels from the original manuscript was an oversight on our part. As shown in our volcano plot, between ad libitum fed and refed mice, there are two significantly *Fos*-associated genes upregulated in the refed group. Therefore, we are confident that the snRNA-Seq analysis accurately captured rapid changes in response to refeeding in the DVC. Only genes differentially expressed (log2 Fold-change >0.5 per group) were considered in the analysis. NS= non-significant.

Author response image 1.



> *(5) The focus on transmitter classification is highlighted, but surprisingly, the well-accepted distinction of GABAergic neurons by Slc32a1 was not used, instead, Gad1 and Gad2 were used as GABAergic markers. While this may be proper for the DMV, given numerous findings that Gad1/2 are not proper markers for GABAergic neurons and often co-expressed in glutamatergic populations, this confound should have been addressed to make a case if and why they would be proper markers in the DMV.*

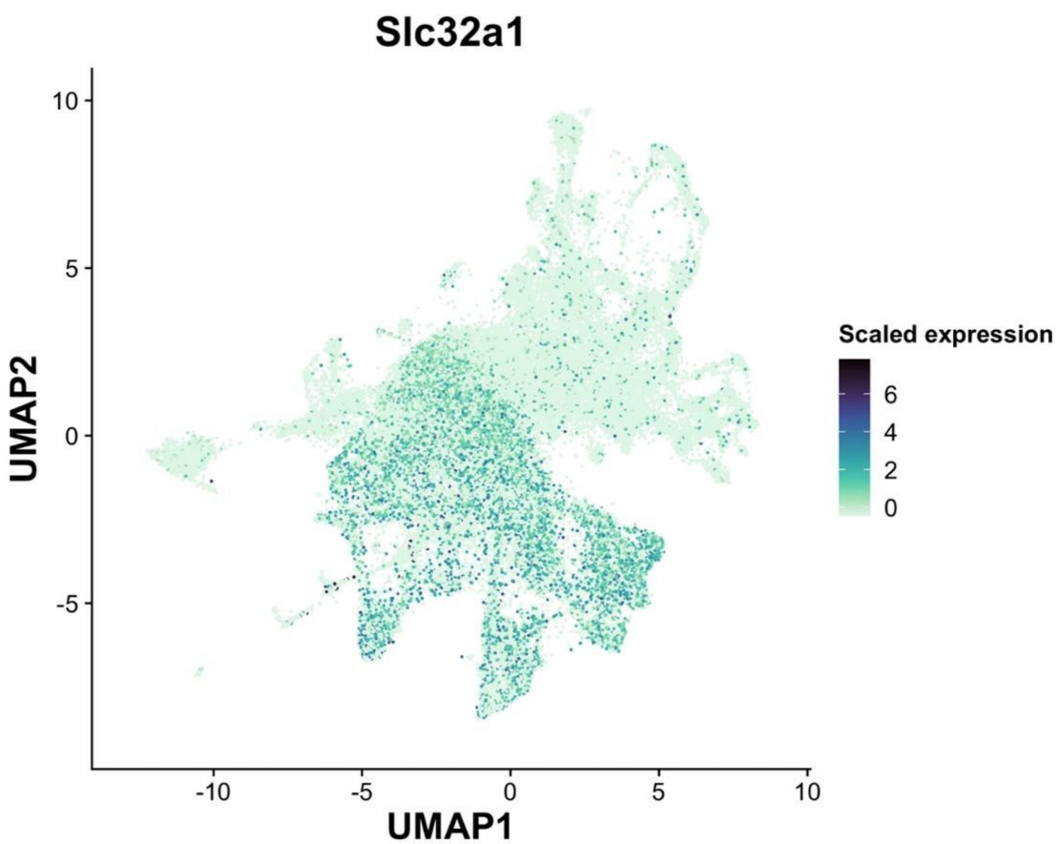
The reviewer raises an important point. Indeed, there are discrepancies in expression between the Gad1/2 genes and Slc32a1 gene in other data sets. To analyze this within our data set, we examined the mainly GABAergic magnaclass 1 (see *Slc32a1* UMAP plot below). In magnaclass 1, only 5% and 3% of all neurons exclusively express solely *Slc32a1* without either *Gad1* or *Gad2*, respectively. In line with the reviewer's comment, we found that 54% of neurons express either *Gad1* or *Gad2* but had no detectable *Slc32a1*. While our failure to detect more cells that co-express *Slc32a1* and Gad genes may be partially due to the low expression of *Slc32a1*, it is also very likely that the DVC, like other brain regions, contains neurons that express the Gad enzymes without co-expression of *Slc32a1*.

This was very much the case with the GLP1 cell cluster, which we identified as the population which had the highest co-expression of excitatory and inhibitory markers. When we refined this analysis to look at expression of excitatory markers with *Slc32a1* (and not other inhibitory genes), there was a marked reduction in the proportion of GLP1 neurons meeting this criterion. We find this is mainly due to the GLP1 cells expressing Gad2 (see plots below). We still find that there are some GLP1-expressing neurons that express excitatory markers and *Slc32a1* and that the GLP1 neurons have a higher proportion of these co-expressing cells than other cell types.

We have extended our results section to reflect this and thank the reviewer for recommending this analysis.

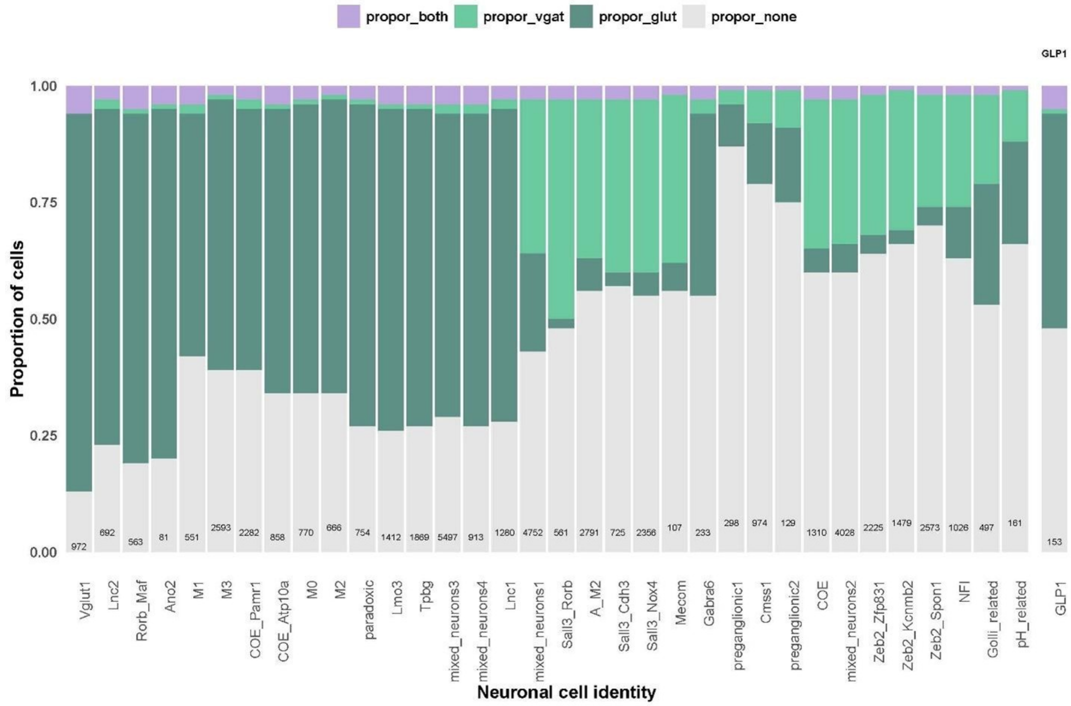
Author response image 2.

Slc32a1 expression across all neurons.



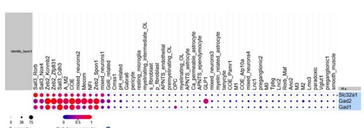
Author response image 3.

Proportion of neurons in all cell identities expressing glutamatergic markers alone (dark green), Slc32a1 alone (light green), both glutamatergic markers and Slc32a1 (purple) or expressing neither Slc32a1 or glutamatergic markers (grey).



Author response image 4.

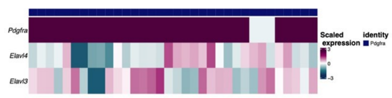
Balloon plot of Slc32a1, Gad1 and Gad2 across cell types. The GLP1-expressing neurons express Gad2 but minimal Slc32a1.



> *(6) The Pdgfra IHC as verification is great, but images are not very convincing in distinguishing the 2 (mouse) or 3 (rat) classes of cells. Why not compare Pdgfra and HuC/D co-localization by IHC and snRNAseq data (using the genes for HuC/D) in the mouse and in the rat? That would also clarify how specific HuC/D is for DMV neurons, or if it may also be expressed in non-neuronal populations.*

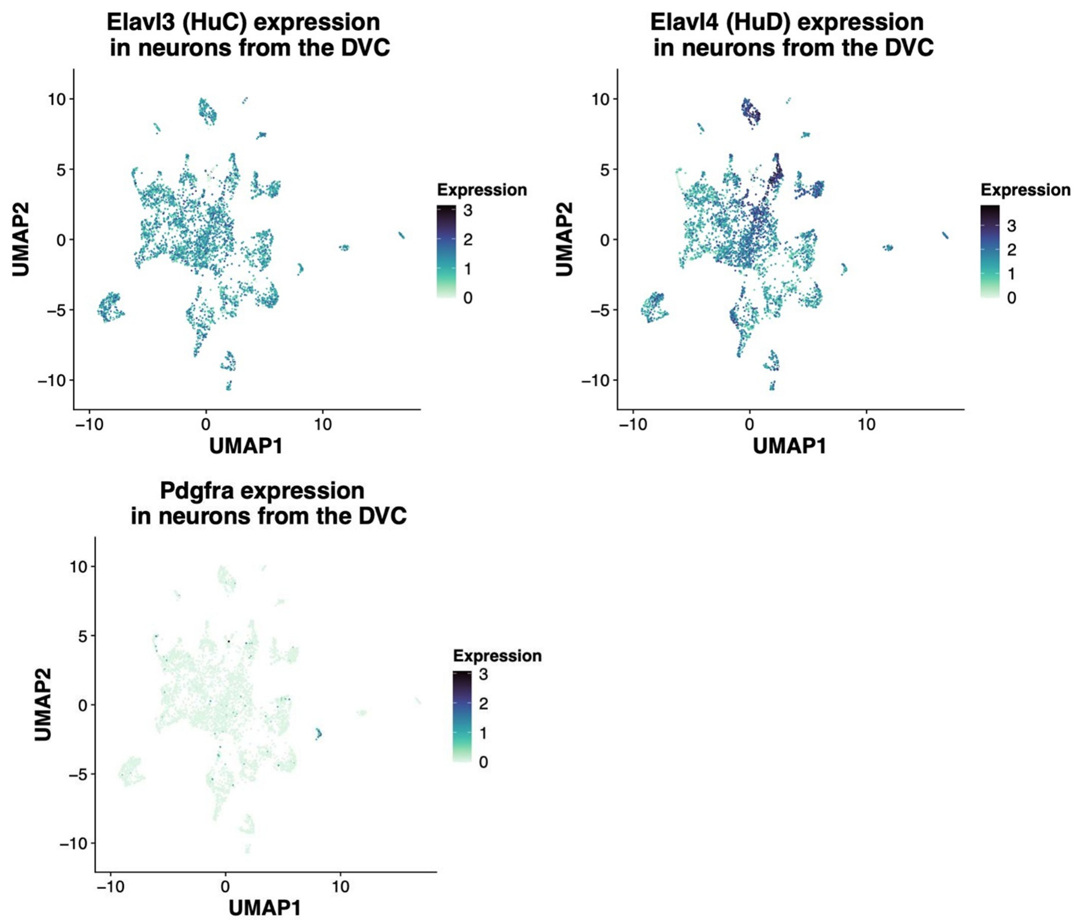
In agreement with the suggestion by the reviewer, we reanalyzed the snRNA-Seq data to identify the extent of the co-expression of HuC/HuD (i.e. *Elavl3* and *Elavl4* genes, respectively) in *Pdgfra*-expressing neurons. The gene expression of the 34 rat neurons belonging to this group are shown in the following heatmap in which each column represents one neuron. As shown, most neurons co-express *Pdgfra* and either HuC or HuD gene. In addition, we shown the UMAP plots of the rat neurons showing expression of the same genes regardless of the neuronal identity assigned. The Pdgfra neurons are visible in darker blue in the last UMAP plot. It's important to note that HuD is a more specific neuronal marker as shown in the table with the average expression of *Elavl3/4* genes, since HuC is expressed by glial cells, specially OPCs and oligodendrocytes. As the HUC/D antibody detects both proteins, this complicates the interpretation of the immunofluorescent staining. While, the snRNA-Seq data suggests these Pdgfra expressing cells are indeed neurons (albeit a rare population), we aim to confirm this in separate studies.

Author response image 5.



Author response image 6.

Average expression (log-normalized counts) of HuC/D by layer 1 cell identity in the rat cells:



Author response table 1.

	connective	glial	neuron	unspecific	vascular
Elavl3 (HuC)	0	2.662964	2.276006	0	0.504195
Elavl4 (HuD)	0.4995	0.375098	5.120204	0.342021	0.723189

> *(7) The importance of sub-clustering for clusters 23, 26, and 27 is not immediately clear. Does this have any relevance to the mouse vs. rat data? Or fed, fast, refeeding data sets? Or is it just to show the depth that can be achieved?*

We appreciate that our justification was not clear within the manuscript. We have clarified our rationale below but briefly, in each case distinct transcriptional profiles were observed, and we pursued this by performing sub-clustering.

Cluster 23 was subclustered as it was found to contain both pre-myelinating and a subset of myelinating oligodendrocytes, therefore, to label them effectively in R instead of cell by cell, those subclusters showing pre-myelinating oligodendrocyte markers were instructed to be labeled as such in the dataset. The remaining cells were labeled as mature oligodendrocytes.

A similar approach was taken for cluster 27 which contained pericytes, endothelial and smooth muscle cells (Figure S5).

In the case of cluster 26, it was possible to find two subclusters of fibroblasts when mapping markers, so they were sub-clustered to instruct in R to label a group with one identity and the other, with the other identity. Therefore, the sub-clustering was done as an aid to label the different identities found through markers mapping (Table S5) in the first clustering round.

All labels were transferred from mouse to rat data using treeArches, including those resulting from the sub-clustering of these clusters. Because this was done to establish identity, it should not be relevant for treatment analyses (e.g. fasted, refed) since they are built from markers that don't change by conditions but remain as identity markers. Indeed, our dataset has an even distribution of these subclusters among samples.

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